

Is There a Disconnection Between CAC 40 Firms and the French Economy? A Geographical Analysis

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Abstract:

The CAC40 Index tracks the 40 largest and most liquid French firms, including some of the world's biggest companies. As with the S and P 500 in the United States, investors use the CAC40 as a reference to measure France's economic strength. However, a striking fact is that the CAC40, excluding banks and insurance companies, generates only 22.9 percent of its revenues in France, while only 50 percent of its international activity now takes place in Europe, compared to 70 percent fifteen years ago, according to EY (2024). Furthermore, almost half of the Index is owned by foreign shareholders (Banque de France, 2025). Therefore, despite being a symbol of France's competitiveness and industrial strength, the Index relies heavily on international activity. This raises the question of how disconnected France's most valuable index may be from the rest of the French economy, and what the main determinants of CAC40 valuation and performance are, national or international, and how these affect companies' financial decisions. Using a panel of 21 companies with yearly data from 2007 to 2024, this paper, considering the challenging macroeconomic environment of the period studied, will seek to determine whether firms' investment decisions exhibit a link through time with France's consumer sentiment, used as a proxy for the underlying French economy. The analysis will also be conducted at the regional level, studying whether activity dependences across markets affect the firms' investment levels and responsiveness to the French underlying economy.

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1 - INTRODUCTION

Since June 2024 when President Emmanuel Macron called snap elections following a defeat to the far right in European elections, France has been at the center of a political crisis having significant economic and social repercussions. The Parliament was left paralyzed, divided in three main camps, making governance of the country very challenging and explaining why there have been three different prime ministers since, a record in the country's history. At the end of 2024, the government was unable to pass a budget for the year 2025, which has had to be based on that of 2024, indexed on inflation, despite the need to cut spending as the fiscal deficit reached 5.7 percent, far past the European Union limit of 3 percent. Because of that macro weakness, France is facing a debt crisis as the country's debt to GDP stands at more than 110 percent of GDP since the Covid crisis, and the country's yields reach those of southern countries, historically considered as riskier (The Economist, September 2025). The CAC 40 Index also seems to have been affected by this crisis. As seen in EY's Profile Report of the Index in 2024, after the hit inflicted by the Covid Pandemic where the Index's revenues decreased by almost 15 percent, and two years of high growth as the economy rebooted, revenue increased by 20.6 percent in 2021 and 22.1 percent in 2022. In 2023 and 2024, revenues decreased by 0.7 percent and 3.3 percent respectively. Similarly, in 2024, investments decreased by 2 percent for the Index. However, given the level of international activity of these firms, what share of that revenue and investment decrease can be attributed to France's economic and political state? Are other countries experiencing similar crises that instead drive those losses for the Index?

For instance, one of the driving questions guiding this research was the S&P 500 Index, the United States' five hundred largest companies by market capitalization, showing signs of disconnection with the underlying economy, to which we refer when looking at indicators such as unemployment, consumer confidence, or performance of small firms in an economy. Indeed, in the United States, in the year 2025, the market continuously reached all-time highs as technology investment, including artificial intelligence, drove growth, despite the levels of uncertainty brought by the Trump Administration's trade war. But what we call the underlying economy has been severely impacted. The job market weakened, forcing the Federal Reserve to engage in a path of interest rate cutting. Unlike the S&P 500, however, the CAC 40 Index's company weights and growth are not overly represented by technology firms that have experienced exponential growth rates in recent years. The four leading companies of the CAC 40 Index in weight, in September 2025, and represented in this dataset as seen in table 1, are Schneider Electric, LVMH, Airbus, and Safran, at 7.74 percent, 7.35 percent, 6.62 percent, and 6.10 percent respectively as reported by Euronext, all from the industrials sector apart from LVMH whose sector is consumer discretionary. Therefore, the CAC 40 is very different from the S&P 500 in the sectors that are represented as the main drivers of growth and given the fact that the United States Index generated only around 30 percent of its revenue from abroad according to S&P Global, studying the French Index will allow us to determine whether that international component drifts away from the French underlying economy.

This topic has many important implications. First, it is well known that companies such as those of the luxury sector, including Louis Vuitton, Hermes, and Kering, are highly dependent on Asian demand for the goods they offer, so studying the relationship with the countries from which the consumers come is crucial. Furthermore, in a context of high geopolitical instability, whether it is in terms of trade under the Trump second term, which has increased tariffs on the European Union, or in the ongoing conflicts in the world, identifying risks and dependences on foreign markets is an essential tool for mitigating social effects and for firms to take the right decisions. Finally, going back to the political crisis described above, in the budget discussions for the year 2025 happening in November and on which the divided Parliament cannot agree, the CAC has been at the center of those discussions. Gabriel Zucman, a prominent economist from the University of Berkeley who will be further discussed in the Literature Review for his work on multinationals shifting profit abroad for taxation purposes, has

proposed a two percent tax on the highest income earners in France, including the operating directors of CAC 40 firms, who have rallied against the decision. This last factor also shows that many factors determine what firms' capital allocations are like.

The Literature Review to conduct this study will thus focus on correlations across markets to gain a better understanding of how the international share of the Index will impact the French Index and could also have implications for the whole economy, as well as on the reasons why firms decide to operate and invest abroad, notably because of their scale and potential tax advantages.

Through the Dataset presented in the next section and the Methodology combining regression analysis and tools such as Principal Component Analysis, after examining the macroeconomic context of the period covered, this study will focus on explaining companies' investment levels when analyzing the respective shares of revenues generated around the world, to not only identify the relationship with the underlying French economy, but also to identify trends and dependences created by different regions.

2 - LITERATURE REVIEW

In *Volatility and Cross-Correlation Across Major Stock Markets* (Ramchand and Susmel, *Journal of Empirical Finance*, 1998), the authors review existing literature on correlation and volatility across markets, showing evidence that correlations indeed exist, vary through time, and tend to increase during periods of high volatility. This last finding will be relevant this study since, as described in the data section, covering the years 2007 to 2024, marked by times of heightened volatility and market co-movement such as the Global Financial Crisis, the European Debt Crisis, and the COVID-19 pandemic. This suggests that to explain movements in the CAC 40, it will be necessary to study its relationship with other foreign markets. This then led to Brodocian's *Analysis of the CAC 40 National Index and the Euronext 100 Pan-European Index Using ARCH Models* (IECS, 2015), which employs various GARCH models, confirming, as in the previous paper, that such tools are well-suited for correlation analysis. Covering 2005–2015, a decade mostly included in this dataset, the author finds strong correlation between the two indices. The EGARCH model, which accounts for heteroskedasticity, successfully explains both markets and reveals higher risk for the CAC 40 due to its lower diversification. Thus it will be possible to compare the results of this study with those of Brodocian, attempting to change the objective to analyzing investment decisions across different world regions.

When reviewing the literature and considering multinational activity, two main reasons for international presence emerge. First, CAC 40 companies are so large and competitive that they cannot confine themselves to the relatively small French market. Their growth naturally drives them to expand internationally, especially toward fast-growing emerging economies. A key study in this area is that of Desai, Foley, and Hines (2009, *American Economic Journal: Economic Policy*) which finds that increasing foreign revenues of U.S. multinationals, controlling for GDP growth, also boost domestic investment. This could indicate a positive impact of foreign activity. However, the study has been criticized for lacking clear causality and for overlooking the influence of tax-haven advantages, which is the second major driver of foreign expansion. Indeed, multinationals often shift profits across borders by underpricing exports to low-tax subsidiaries, paying intra-group interest, or relocating intangible assets. Gabriel Zucman, a leading economist in this field (and a central figure in the French tax debate advocating a 2% wealth tax on top incomes, including CAC 40 executives as mentioned above), estimates in a 2021 WIDER paper that by 2015, as much as 36% of global multinational profits were shifted to tax havens, leading to a 10% loss in global corporate tax revenues. Complementing these findings, Keightley's 2013 report to the U.S. Congress shows that American multinationals reported

disproportionate profits in low-tax countries such as Ireland and Bermuda relative to their real operations there, while reporting less in higher-tax countries like Mexico or Canada.

Finally, this study also draws on *How Sovereign Is Sovereign Credit Risk?* (Longstaff et al., 2011), which uses principal component analysis to show that 64% of the variation in sovereign CDS spreads across more than 20 countries is explained by U.S. and global market factors rather than domestic ones. Inspired by this approach, to account for the multicollinearity that might exist between different international markets, not including those from France and Europe, the factor augmented methodology will be used to determine which weight to assign to each international market when studying dependence on external factors while controlling for share of production in France or in Europe.

Lastly, following the 2022 war in Ukraine, which profoundly affected global markets through rising geopolitical risk, Caldara et al.'s (2022) *Measuring Geopolitical Risk* study finds evidence of the influence of such geopolitical shocks on Indices such as the CAC 40, and will thus be significant to cover the dataset period and the geopolitical risk it included throughout.

3 – DATA & SUMMARY STATISTICS

Out of the forty CAC 40 Companies in 2024, this study will includes 21 companies, including 18 yearly observations across time for each company, from 2007 to 2024. Because of their role played in the economy not only as profit generating entities, but also as key financial intermediaries, and as suggested by the Literature Review, the four banks AXA, BNP-Paribas, Credit Agricole, and Societe Generale were purportedly left out of the dataset. Then, fifteen out of the remaining thirty-six companies had unfortunately to be left out because of a lack of data, especially for the share of revenue generated abroad. Table 1 below shows the 21 firms used for this analysis, describing their sector and the weight they represented in the Index as of September 2025.

Table 1: CAC 40 companies included in the dataset, sector and weight in the index, September 2025

Company	Sector	Weight (%)
Schneider Electric	Industrials	7.74
LVMH	Consumer Discretionary	7.35
Airbus	Industrials	6.62
Safran	Industrials	6.1
Air Liquide	Basic Materials	5.78
Sanofi	Health Care	5.17
L'Oreal	Consumer Discretionary	4.99
Hermes	Consumer Discretionary	3.72
Danone	Consumer Staples	2.71
Legrand	Industrials	2.08
Engie	Utilities	1.88
Michelin	Consumer Discretionary	1.22
Kering	Consumer Discretionary	1.18
Dassault Systemes	Technology	1.08
Capgemini	Technology	1.02
Pernod Ricard	Consumer Staples	0.95
ArcelorMittal	Basic Materials	0.74
Carrefour	Consumer Staples	0.46

Bouygues	Industrials	0.41
Renault	Consumer Discretionary	0.38
Edenred	Industrials	0.27

Source: EuroNext, September 2025

The data has been collected from Bloomberg, the World Bank Open data source, the French National Bureau of Statistics (INSEE), and FRED, including, for the purpose of this study four different categories listed in the table below, where the company specifics will be unique to each company at each time period while all other yearly observations will be common to all entities, totalling more than 500 observations.

Table 2: Data by Category and by Source, 2007-2024 & 21 companies

	Company Specifics	Global Factors	French factors	European factors
Bloomberg	Stock price, Total Assets, Absolute Capex, Net Income, Revenue (Growth), Debt to Equity ratio, Market Capitalization, Revenue Share by Location	VIX Index, MSCI EM Index, MSCI Asia Index, S&P, CRB Commodity Index, DAX, FTSE, India NIFTY Index, IBOVESPA Index, Nikkei 225, Shanghai Composite Index	CAC 40 Index, Small CAC Index, Mid CAC Index, 10year yield, CDS spread	MSCI EU Index, EU VIX, EUR/USD
World Bank Open Data			French GDP growth, FDI as percent of GDP	
INSEE			Unemployment	
FRED			French Consumer Sentiment	ECB Fed rate

Table 3: Summary Statistics for Air Liquide and its Company-specific category

Variable	Obs	Mean	Std. Dev.	Min	Max
StockPrice	18	98.04	37.303	47.071	177.145
TotalAssets	18	40862.479	10578.051	26674.785	53677.676
Capex	18	2732.498	580.268	1863.326	3813.375
NetIncome	18	2366.662	571.082	1539.544	3576.465
RevenueGrowth	18	5.531	9.255	-8.601	28.28
MarketCap	18	5.417e+10	2.314e+10	2.383e+10	1.021e+11
TotalDebttoEquity	18	71.777	15.189	45.067	98.767

**where TotalAssets, Capex, NetIncome are in millions of US dollars, and StockPrice, MarketCap are in US dollars*

Table 4: International Markets Indicators Summary Statistics

Variable	Obs	Mean	Std. Dev.	Min	Max
SP500	18	2613.479	1457.129	903.25	5881.63
NIKKEI	18	174.091	54.032	98.08	265.803
ShanghaiComposite	18	456.075	103.85	266.765	720.357
NIFTY	18	155.891	60.155	60.794	276.294
MSCIEmergingMarkets	18	1019.902	175.435	567.04	1291.26
MSCIAsiaexclJapan	18	594.066	129.159	288.11	842.93
VIX	18	19.456	6.569	11.04	40

Table 5: European Markets Indicators Summary Statistics

Variable	Obs	Mean	Std. Dev.	Min	Max
DAXIndex	18	13025.03	3903.934	6789.116	20693.561
UKIndex	18	9542.175	1304.954	6462.803	12802.096
EuropeanVIX	18	27.162	11.223	15.019	61.914
MSCIEurope	18	146.11	25.458	96.875	185.831
ECBDFR	18	.605	1.424	-.5	3.73
EURUSD	18	1.222	.138	1.035	1.459
EUconsumerSentiment	18	-12.539	5.663	-22.1	-2.5

Table 6: French Economy Indicators Summary Statistics

Variable	Obs	Mean	Std. Dev.	Min	Max
CAC40Index	18	6107.268	1340.096	4095.114	8345.774
smallCAC	18	11104.074	3090.739	5182.632	16957.307
French10yearYield	18	1.963	1.463	-.338	4.422
FranceCDS	17	49.62	51.176	8.059	222.296
GDPgrowth	18	1.016	2.788	-7.441	6.882
Unemployment	18	8.789	1.087	7.308	10.354
Inflation	18	1.655	1.459	.038	5.222
France consumer sentiment	18	-12.944	4.235	-18.5	-5.8

One of the foundations of our analysis is the share of revenue generated by Location for each company across the years. Because of the difficulty to get this data either from Bloomberg or company yearly reports since it is not always available, even for the selected 21 companies the data is more unbalanced than the others as sometimes data is missing for early years, and very importantly, companies do not report in the same for each region, meaning that for some companies we will have the share only in Europe, while for others only in France, and for others in both, but also in the different continents such as Latin America, North America, or Asia, companies will not report homogeneously depending on where

they produce. The table, below shows for which region companies reported their share of revenue, and Appendix 1 details specifically for each company the share of revenue in different location across time, as well as presents a pie chart with that revenue share in 2024.

Table 6: Revenue Share by Location, as reported

France	Europe	North America	Asia Pacific	LATAM	Japan	Africa / Middle East
Airliquide Bouygues Capegemini Carrefour ENGIE LVMH SAFRAN	ALL, except : Engie Bouygues	ALL, except: Bouygues Carrefour 'Dassault Sytemes' Edenred Hermes Renault Sanofi	Air Liquide Airbus Danone Dassault Systemes Hermes Kering LVMH Pernod Ricard 'Schneider Electric' Safran	Airbus, Carrefour Danone* Edenred Engie, L'Oreal	Kering, LVMH	AIRBUS Bureau veritas*, Safran*

**indicates missing data for first periods - To see company by company data, refer to Appendix I*

This table and Appendix 6 thus stand as the pillar of our analysis to make key observations as well as to study different trends when companies operate in a single region, and to conduct an analysis at the regional level too.

Finally, as further discussed in 9 –*Limitations and Further Research Ideas*, it would have been optimal – and it will thus be discussed qualitatively – to present two additional tables including companies in which the French government (or others) owns a share, as well as one with the tax advantages offered by different regions and variables which are not accounted for in this dataset and that could be explanatory.

4 - METHODOLOGY

This Paper will work with the statistical properties of Panel Data for Regression Analysis and will also use methodologies Principal Component Analysis to account for multicollinearity in markets. The Section on *Limitations and Further Research Ideas* will discuss Further Research Ideas and applications for the dataset and in answering similar questions enabled by time-series data, using models such as the Auto-Regressive Integrated Moving Average (ARIMA), or Arellano Bond.

Panel data fixed effects

The methodology will make use of the fixed effects for each entity as is done in Panel Data and which in this case would represent the determinants not changing through times across firms, such as the company's 'culture' or norms across the industry in which it operates. The possibility of replacing that

fixed effect by dummy variables not changing through time for each sector (for example for Consumer Discretionary or Industrials), as seen in Table 1 has been evaluated, however because there are only 21 entities and these are multinational firms operating in different regions, and sometimes across different sectors, it is believed that a fixed effect for each entity works best.

The inclusion of time fixed effects through dummy variables for each period will not be included in the regression analysis. Despite it being useful to account for effects affecting all companies in a given year, which is necessary for the period covered that includes the 2008 Financial Recession, the Euro credit crisis, the COVID Pandemic, the invasion of Crimea in 2014 and of all of Ukraine in 2022, or even the 2024 political crisis in France, we will use variables in the dataset as proxies for these events. These include the VIX when studying events with international repercussions, and French CDS spreads for example to account for French-specific risks. This will then enable us to single out and evaluate the effects of more specific factors, such as the performance of an index in each country, which would otherwise be absorbed by these dummy variables.

Working with time series

When evaluating Asset Prices to evaluate a company's performance, reflected here in the Stock Price of the CAC 40 companies, or even studying the level of investing from companies, one of the first variables that comes to mind in determining that number is the lagged value. Therefore, in our regression, a lagged term will be included, which will end up always being significant. Because of the inclusion of the lagged term, however it will be important to account for the Nickell bias that might arise as the lagged dependent variable will also include its error term from regression at the previous period. Thus, to ensure that robust inference can be drawn it will be important to study the autocorrelation across error terms that might bias the coefficients obtained for the explanatory variables.

Additional Tools

In many of the regressions a time dummy will be included to characterize the first nine years of the data and another one for the second half, which when interacted with other variables, allow for inference across time, for instance when studying if the French indicators like Consumer Confidence, or even to estimate if certain regions are significant in explaining changes in the dependent variable.

Finally, in some cases, especially when wanting to estimate the effect of various markets in a certain region, Principal Component Analysis (PCA) will be applied for Factor Augmented Research, ensuring that the first component is high enough, and placing that first component as a replacement of the various markets. This will, for instance, be applied when working with companies who report their revenues in France (Table 6), using PCA for markets outside of Europe, and when examining the companies working in Asia, to study the different markets that are present.

Acknowledging these properties drawn from our data, as well as the strengths and limitations of it, we proceed to the regression analysis.

Regression Analysis

The proposed Regression Analysis is the following where the set of variables are described below in 2 different equations that can yield different combinations when using additional tools such as PCA.

$$\frac{Capex_{i,t}}{Assets_{i,t}} = \frac{Capex_{i,t-1}}{Assets_{i,t-1}} + \alpha_i + \Gamma'X_{i,t} + FinancialCrisis_{2009} + \beta_G G_t + \beta_F F_t + Region_{i,t} + u_{i,t} \quad (1)$$

$$\frac{Capex_{i,t}}{Assets_{i,t}} = \frac{Capex_{i,t-1}}{Assets_{i,t-1}} + \alpha_i + \Gamma' X_{i,t} + FinancialCrisis_{2009} + \sum_{i=1}^2 \beta_{G,i}(G_t \times Period_{t=i}) + \sum_{j=1}^2 \beta_{F,j}(F_t \times Period_{t=j}) + Region_{i,t} + u_{i,t} \quad (2)$$

From Equation (1):

$\frac{Capex_{i,t}}{Assets_{i,t}}$: The dependent variable represents the investment ratio of the a given company in each period and the first regressor is its lag.

α_i : entity fixed effects

$\Gamma' X_{i,t}$: Company specific vector of explanatory variables, such as Net Income, or Debt to Equity ratio

FinancialCrisis₂₀₀₉: Dummy variable for the year 2009 when companies experienced huge decreases in Capex because of the Financial crisis

$\beta_G G_t$: Vector of International Indicators, such as Global Markets or the VIX

$\beta_F F_t$: Vector of French Indicators

Region_{i,t}: Chosen Region to control for depending on the analysis

$u_{i,t}$: Error terms

From Equation (2): One of the key components of this research is not only to evaluate the level of disconnection that could exist between the French economy and the CAC 40, but also to determine whether this is a relationship that has changed through time. Thus, we will employ a model of regression for which the time period changes through the dummy variable **Period_t**, that when set to **Period₁** will represent the periods from 2007 to 2015, and **Period₂** representing the periods from 2016 to 2024

$\sum_{i=1}^2 \beta_{G,i}(G_t \times Period_{t=i})$: In the analysis this represents the Global Variable interacted with a dummy variable used to represent a certain period, so that it can capture an effect that might change through time when both are included in the regression

$\sum_{j=1}^2 \beta_{F,j}(F_t \times Period_{t=j})$: Similarly, this represents the French Indicators interacted

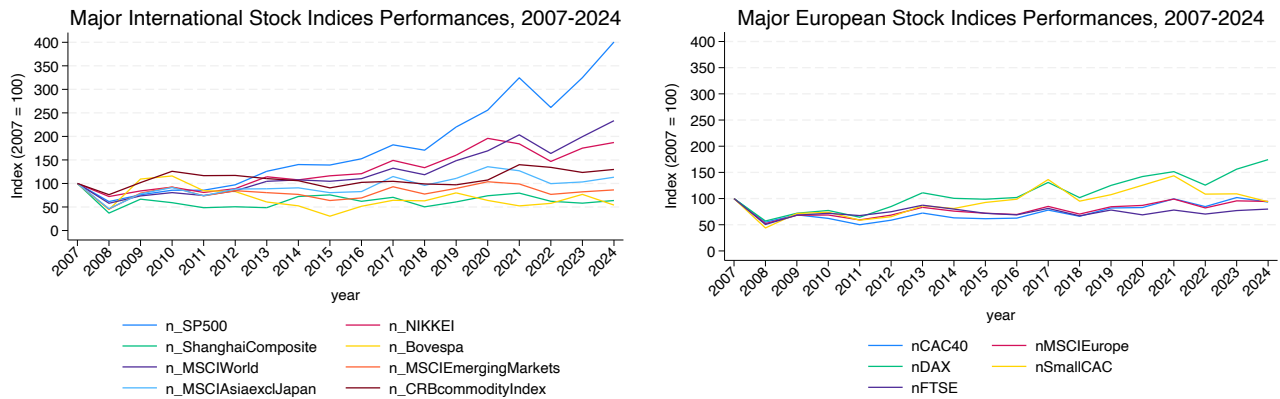
As in both cases the sum iterates over two periods in equation (2), the dataset will be split into 2 at the halfpoint being the transition between the years 2015 to 2016 as described above, and therefore, since we're applying this methodology, before running any of the regressions it will be important to study the macroeconomic environment of the two periods.

Replacing $\beta_G G_t$ by $\beta_{PCA} PC1_t$ in (1) or (2) leads to the factor augmented analysis where we're accounting for correlation between different markets, for example, as seen in the subsequent sections.

5 – MACROECONOMIC ENVIRONMENT OF THE PERIOD 2007 - 2024

Studying the trends from the Macroeconomic environment at the French and International level is key determining which of the controls or explanatory variables should be used in the regression analysis, as well as checking the different relationships that might exist between the variables, to then interpret the regression output in a more precise way.

Figure 1: International & European Markets Performance, Index (2007 = 100)



From Figure 1 and restricting the macroeconomic analysis to Market Indices, it is first possible to quantify some of the impacts of the Great Financial Recession in Europe. The hit was huge as compared to the 100 level in 2008, it decreased to less than 50, on both graphs. What is interesting is the speed at which different market indices recovered to the 2007 level: for many of the international indices on the left, they seemed to have achieved that around the years 2014 (except S&P500 which recovered before and the BOVESPA or Emerging Markets that remained below 2007 level) while in Europe, apart from the DAX, which recovered in around 2013 and remained at that level until 2018, other National Indices such as the CAC 40 or the FTSE only grew at a small pace and haven't yet recovered to 2007 levels. This illustrates one key macroeconomic trend of the past 18 years, which is the slow growth of the European continent compared to North America or Asia, as we see from Figure 1, and which could have important implications for answering our research question. Indeed, if CAC 40 companies are more exposed to foreign markets which exhibit higher growth, then it could be possible that they would benefit from these levels from abroad as compared to firms in France only operating at the local or national level. However, another surprising feature from this figure, is that the small CAC index which represents 90 smaller listed companies in France grew at a much faster rate than the CAC 40, even though it converged back to the same level in recent years. This is surprising since we would have expected the opposite relationship, and it is possible that because the small CAC companies remain large, they could very well have built resilience and benefit from higher growths from abroad, for example. This thus makes it difficult to use it as our French indicator explanatory variable since the correlation between the small CAC and the S&P500 for the period stood at 0.65.

Figure 2: The French Macroeconomic Environment



From Figure 2, looking at the GDP growth, we see that France's economy matches the observations for Figure 1, as the country took a hit from the Great Financial Recession, and was then indeed characterized by low levels of growths as seen in the European indices converging back to Index equal to 100. The Covid 19 crisis was then also an element to consider making GDP growth dip even more, to almost -8% and the period that followed was a temporal bounce back to historical levels of growth at 8% before converging back to low growth today, coinciding with the challenges around the Fiscal Deficit described in the Introduction. From looking at the unemployment graph, it is very insightful if as described in the methodology section a first dummy will be used for the periods 2007 to 2015 and 2016 to 2024. The unemployment graph demonstrates two different behaviors characterizing these periods, in the former a slow increase each year, which is surely linked to the Euro debt crisis which arose in the beginning of the second decade of the 21st century, while in the latter decreasing unemployment suggested that the economy was becoming more resilient. The French 10 year graph also reflects those macroeconomic trends as lower rates following the 2008 crisis aimed to restore economic activity, and could have helped the investment ratios of companies by lowering their cost of financing, so it is a variable we'll include in our regression analysis. With the Covid crisis rates were brought to the zero lower bound ensuring a strong bounce back following the crisis, but because of the high inflation and the risks mentioned above rates are today at their highest which could hurt the economy. Finally, the exchange rate with the dollar was included because of the role played by currency valuation for firms exporting goods, and the constant depreciation of the euro for the period (with the Euro debt crisis as a key reason) could be a factor helping CAC 40's exports, making their goods more competitive – but that could of course also raise their costs.

Figure 3: Consumer Sentiment, the French Indicator for the Analysis

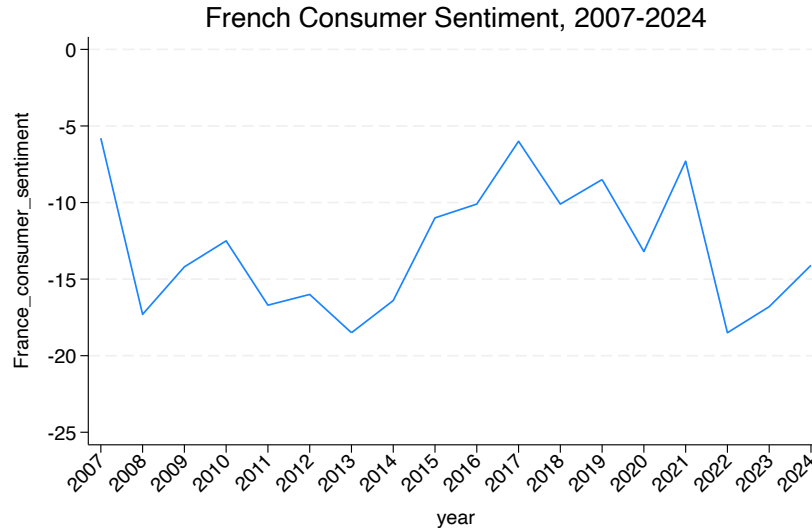


Table 7:

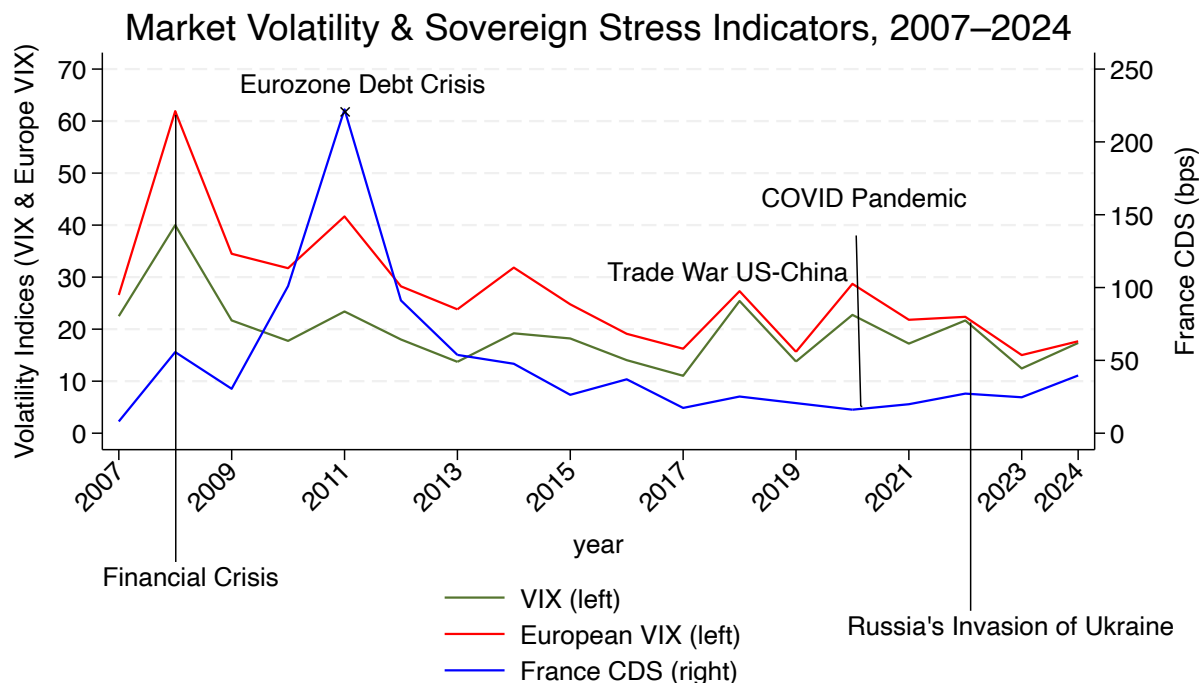
French Consumer Sentiment, CAC40, S&P 500

Correlations

Variables	(1)	(2)	(3)	(4)
(1) France_consumer_sentiment	1.000			
(2) CAC40Index	0.331	1.000		
(3) SP500	0.086	0.749	1.000	
(4) GDPgrowth	0.287	0.179	0.128	1.000

From Figure 3, Consumer Sentiment is identified to be the variable reflecting the macro trends described above (Financial crisis, boom in the second decade of the century until the impacts of Covid), and most importantly, it has the feature of bringing economic relevance by interesting itself to the French consumer, while also being correlated to the CAC 40 and capturing a relationship that is not linked to International Markets such as the S&P 500, as seen by the correlation at almost 0, presenting another side of the French economy. Therefore, we will use this variable in the regression analysis to see whether when investment in different regions is regression on it, some of the effect is still captured, or if firms' decisions are independent from it, studying the sign and significance of the yielded coefficients.

Figure 4: Measures of risk and key events



The VIX Index will be used throughout the analysis as a way of controlling for the events shown in Figure 4 that would have impacted all companies.

In *Appendix 4* the tables of correlations among international markets can be found.

6 - COMPANIES WHO HAVE REPORTED DATA FOR FRANCE

As seen in Table 6, 6 out of 21 of our companies from our data sample report precise revenue shares in France: Air Liquide, Bouygues, Capgemini, Carrefour, Engie, LVMH, Safran. The evolution of their share of revenue since 2007, as well as those numbers in 2024, and the Stock Price, and Investment Levels are available in Appendix 1. These companies represent the most relevant and precise datapoints for answering the question this study asks as these companies reporting these numbers is already a sign that much of their activity depends on France, as compared with other companies for which we have the European number, but no indication of the share represented by France. For instance, from Appendix 1, Bouygues is the company with the highest total share in France across the years and in 2024, with over 70% of revenue generated in France in 2007 and in 2024 that number fell to almost 48%, but it is still the highest value.

After using the Principal Component Analysis framework, we will thus interest ourselves to the Investment Ratios of these firms, based on the French and International Macroeconomic environment introduced earlier.

Effectiveness of Principal Component Analysis and the data representative of the French Economy to draw inference

To determine the impact that international factors, such as the performance of various countries' indices in determining levels of investment across firms, the Principal Component Analysis framework will be applied international markets that not include France nor Europe (apart from MSCI World) in the objective of isolating the effect that would otherwise have been captured in indices including France. As seen in the table below, the first augmented factor variables represents 60% of the variance, with almost all indices moving together – apart from Brazil, which would be a problem if working with firms generating much of their revenues there, such as ArcelorMittal, or in Latin America. Given that in the following samples dependence on France and on the rest of the world without differentiating between countries will be studied, this framework is convenient.

Figure 5: Principal Component Analysis, International Markets excluding Europe

Principal components/correlation	Number of obs	=	378
	Number of comp.	=	10
	Trace	=	10
Rotation: (unrotated = principal)	Rho	=	1.0000

Component	Eigenvalue	Proportion
Comp1	6.252	0.625
Comp2	2.130	0.213
Comp3	0.830	0.083
Comp4	0.400	0.040
Comp5	0.319	0.032
Comp6	0.037	0.004
Comp7	0.018	0.002
Comp8	0.009	0.001
Comp9	0.003	0.000
Comp10	0.000	0.000

Variable	Comp1	Comp2	Comp3	Comp4	Comp5	Comp6
NIFTY	0.373	-0.084	0.165	0.233	0.420	-0.330
ShanghaiCo~e	0.198	0.398	-0.559	-0.338	0.589	0.115
Bovespa	-0.063	0.514	0.661	0.180	0.312	0.350
MSCIEmergi~s	0.275	0.442	0.224	-0.282	-0.294	-0.390
MSCIWorld	0.380	-0.191	0.063	0.153	0.123	-0.120
MSCIAsiaex~n	0.367	0.147	0.092	-0.409	-0.318	-0.115
UKIndex	0.191	0.457	-0.382	0.649	-0.344	0.047
DAXIndex	0.387	-0.128	-0.039	0.170	-0.118	0.360
NIKKEI	0.377	-0.173	0.068	-0.230	-0.134	0.637
SP500	0.366	-0.248	0.093	0.160	0.161	-0.194

Principal components (eigenvectors)

Components 7 to 10 have been ignored for presentation purposes

Table 8: Matrix of correlations – Small CAC, CAC 40 and PC1

Variables	(1)	(2)	(3)	(4)	(5)	(6)	(7)
(1) smallCAC	1.000						
(2) CAC40Index	0.722	1.000					
(3) PC1	0.842	0.859	1.000				
(4) smallCACxPe~2015	-0.584	-0.398	-0.589	1.000			
(5) smallCACxPe~2024	0.835	0.587	0.769	-0.934	1.000		
(6) PC1xPeriod_~2015	0.818	0.691	0.813	-0.409	0.637	1.000	
(7) PC1xPeriod_~2024	0.595	0.737	0.850	-0.563	0.643	0.384	1.000

*where the variables followed by a date indicate that the methodology of separating period has been applied, i.e. creating one period from 2007 to 2015, and one from 2016 to 2024

Table 9: Matrix of correlations – France Consumer sentiment, CAC 40 and PC1

Variables	(1)	(2)	(3)	(4)	(5)	(6)
(1) France_consumer sentiment	1.000					
(2) PC1	0.324	1.000				
(3) consumer_se~2015	0.525	0.782	1.000			
(4) PC1xPeriod_~2015	0.482	0.813	0.758	1.000		
(5) consumer_se~2024	0.019	-0.712	-0.841	-0.584	1.000	
(6) PC1xPeriod_~2024	0.077	0.850	0.553	0.384	-0.601	1.000

From Table 8 and 9 it can be seen see that for this section, the Consumer Sentiment is much more isolated from the international markets factored in the first component, since it exhibits much lower levels of correlation for the period, 0.32, as compared to the correlation of almost 0.85 between the Small CAC Index and the First Component, so we when using Small CAC without any other International Index explanatory variable, it will be anticipated that it will account for some of the changes in those indicators too.

Following the Methodology detailed previously, the regression Analysis builds as follows:

Table 10: Regression for Companies reporting Revenue Share in France

VARIABLES	(1) Control Factors with ENGIE	(2) Control Factors without ENGIE	(3) SmallCAC Period1	(4) SmallCAC Period2	(5) PCA Period1	(6) PCA Period2	(7) sentiment Period1	(8) sentiment Period 2
L.InvestmentRatio	0.481*** (0.093)	0.502*** (0.101)	0.429*** (0.069)	0.380*** (0.072)	0.437*** (0.087)	0.440** (0.138)	0.399*** (0.059)	0.421*** (0.088)
NetIncome	0.000*** (0.000)	0.000*** (0.000)	0.000*** (0.000)	0.000*** (0.000)	0.000*** (0.000)	0.000*** (0.000)	0.000*** (0.000)	0.000*** (0.000)
FinancialCrisis2009	- 0.881*** (0.219)	- 0.956*** (0.226)	-1.267*** (0.287)	-1.417*** (0.268)	- 1.269*** (0.278)	- 0.975*** (0.202)	- 1.338*** (0.303)	- 1.155*** (0.259)

ratesXperiod_2010_2015	0.079 (0.060)	0.122* (0.051)	-0.048 (0.070)	-0.142* (0.064)	-0.015 (0.021)	0.076 (0.081)	-0.128* (0.060)	-0.016 (0.078)
VIX	0.032* (0.014)	0.036* (0.017)	0.031 (0.017)	0.013 (0.016)	-0.005 (0.010)	0.032 (0.019)	0.014 (0.013)	0.031 (0.018)
France	0.018 (0.021)	-0.003 (0.015)	-0.010 (0.016)	-0.009 (0.014)	-0.017 (0.020)	-0.010 (0.009)	-0.012 (0.015)	-0.025 (0.017)
smallCACxPeriod_2007_2015			0.000** (0.000)					
smallCACxPeriod_2016_2024				-0.000*** (0.000)				
PC1xPeriod_2007_to_2015					-0.278** (0.077)			
PC1xPeriod_2016_to_2024						-0.101 (0.091)		
consumer_sentiment_2007_2015							-0.053** (0.017)	
consumer_sentiment_2016_2024								0.045* (0.018)
Constant	0.633 (0.456)	1.007** (0.289)	1.454*** (0.335)	2.740*** (0.393)	2.306*** (0.184)	1.632 (0.870)	1.896*** (0.293)	2.413** (0.785)
Observations	119	102	102	102	102	102	102	102
R-squared	0.349	0.339	0.377	0.404	0.384	0.361	0.402	0.386
Number of firmid	7	6	6	6	6	6	6	6
Robust standard errors in parentheses								
*** p<0.01, ** p<0.05, * p<0.1								

ENGIE, which is one of the French firms in the dataset, is included in the first regression, but when looking at its share price and Capex levels in the Appendix 1, it clearly begaves as an outlier (Capex increasing during the crisis and then decreasing throughout the entire period as opposed to the grand majority of other firms), so it is removed from regression (2) on the objective of seeing stronger results.

In (1) and (2) where control factors are experimented, various hypotheses are confirmed such as the lagged value of Investment Ratio being significant, as well as the logic behind the significance of the net income and financial crises variables. The decrease in rates, however, interacted with the period shown above does not seem to be significant in most cases, and experiences coefficient changes, so the point that firms' WACC for example would be lower is not inferred robustly. However, it is still kept throughout the regressions in the objective of controlling for that overall effect in the economy. The VIX is significant at the 10% level which was expected and establishes trust in the regression methodology, while the variable France is meant to control for each firm's exposure to the economy.

Despite our earlier comments about the small CAC variable, we still wish to test it in this first regression and observe that it is very significant, at the 5% then 1% level in the two periods, respectively, where its coefficient is first positive, then negative.

The PCA's first component results are very relevant since they lose in significance in the second period, as shown in (6), which could be explained by the weights it takes not being relevant in the second decade, if they do not reflect the true dependence of the determinants of firm investing, from an international markets perspective.

Finally, the Consumer Sentiment two significant coefficients provide robust foundations behind the reasoning that because these firms have reported revenue share for France, and that that number is high as seen in Appendix 1 and for Bouygues (far ahead of other companies' share in Europe), that consumer sentiment and part of the French economy not accounted for in international markets' correlation remain relevant throughout the period.

7 – COMPANIES REPORTING SHARE OR REVENUE IN EUROPE

This section will include a larger number of entities as 19 out of the 21 companies report their share of revenue in Europe, Bouygues and ENGIE being the excluded ones. We will thus be newly working with 4 of the companies studied in the previous section and will use the share of revenue generated in Europe as a relative proxy for what the level could have been in France.

The 19 firms exhibit a wide range of different revenue distributions across time and at the present in the year 2024, so when pooling them, the aim will be to identify some key trends as well as if there are certain dependences on foreign indicators that stand out. In the previous section it was important to study the effect of the “small” CAC Index using regression analysis – to determine what role its correlations with international markets as well as its stronger performance than the CAC 40 in the second decade of the 21st century had on the level of investing of firms – but here the proxy which will be used for the state of the French economy will solely be consumer confidence. The first Principal Component will also be employed, but now because of the range of continents represented, the target will also be to single out the effect of different market indices across firms.

A few patterns can be observed before digging into the analysis. (1) When looking at those firms, the general observation for the share of production across countries is that it remains relatively constant across regions through time, but we do seem to observe that the share of revenue generated in Europe or in France when available trends downward – see Danone in Appendix 1 for an example and Edenred as a counterexample.

Many of the firms, 16 out of 21, report revenue share from North America, showing the dependence of the Index on the continent, and having important implications going back to the current macroeconomic context presenting challenges in terms of trade and political partnership. This can perhaps be linked to the general trend observed amongst companies, which is one of CAPEX and Stock Price growth throughout the period, where North America may well play a role.

Firms generating a large share of their revenue in Asia, historically and in 2024, such as Dassault Systemes (more than 20% in 2024), as well as all the firms of the luxury sector like Hermès (more than 50% in 2024), Kering (35%), LVMH (30%), Pernod Ricard (43%), Schneider Electric, seem to exhibit a common trend: very high levels of investing in recent years, sometimes starting in the middle of the 2010s decade while other times later. That relationship does not hold for the Stock Price, however, which is seen to take a shock following the Covid pandemic, as is the case for Kering or Pernod Ricard. It is not known if the CAPEX spending is targeted at production linked to the continent, but some of the factors that may play a role would be the resilience or growth of Asian economies, encouraging firms to invest more, linking it back to Figure 1: the Japanese NIKKEI was one of the best performing indices. In this section we will test for firms having a large share of their revenue in Asia having higher investment ratios in the last 8 years, and in the subsequent section on Asian-producing firms, we will seek to determine which of these national indices drive that trend, if it exists.

In contrast, the companies who report slower growth in CAPEX or even Stock Price seem to be dependent on the Latin American region. These include companies which are known to have been in turmoil in recent years, such as Carrefour, ArcelorMittal (discussions about nationalization at the core of

the French economic and political debate now), and Danone. Would the dependence on the continent be an indicator of that limited resilience?

Table 11: Determinants of Investment - Consumer Sentiment				
VARIABLES	(1) Control Factors	(2) PC1	(3) Sentiment	(4) Peiod Sentiment
L.InvestmentRatio	0.652*** (0.096)	0.621*** (0.112)	0.626*** (0.110)	0.622*** (0.112)
NetIncome	0.000** (0.000)	0.000** (0.000)	0.000** (0.000)	0.000** (0.000)
FinancialCrisis2009	-1.092** (0.403)	-1.267** (0.455)	-1.249** (0.449)	-1.252** (0.451)
PC1xPeriod_2007_to_2015		-0.186** (0.067)	-0.209*** (0.058)	-0.185*** (0.060)
PC1xPeriod_2016_to_2024		0.013 (0.034)		
VIX	0.018** (0.009)	-0.004 (0.007)	-0.011 (0.007)	-0.009 (0.007)
Europe	0.002 (0.006)	-0.005 (0.006)	-0.005 (0.006)	-0.005 (0.006)
ratesXperiod_2010_2015	0.065 (0.055)			
MSCIWorld_2016to2024			0.000 (0.000)	
France_consumer_sentiment			-0.030*** (0.010)	
consumer_sentiment_2007_2015				-0.026** (0.011)
consumer_sentiment_2016_2024				-0.033*** (0.011)
Constant	0.435* (0.231)	1.086** (0.412)	0.690 (0.458)	0.801* (0.408)
Observations	275	275	275	275
R-squared	0.419	0.439	0.457	0.453
Number of firmid	19	19	19	19
Robust standard errors in parentheses				
*** p<0.01, ** p<0.05, * p<0.1				

PC1, for the period 2007 to 2015, is significant at least at the 5% level in different cases, as a proof that following the financial recession and reaching the European debt crisis, companies must have moved together in the decreased level of investment which would slowly start to pick back up, while for the second 9 years, the non-significant coefficient was dropped. Results are significant since the level of investment in (3) is highly linked to French consumer confidence over the period, even after accounting for the movements of markets as well as risk indicators, and the two-step process followed in (3) and (4) indicates a robust relationship.

Table 12: Determinants of Investment - Regional Analysis				
VARIABLES	(1) Asia	(2) Asia and N.A.	(3) North America	(4) Latin America
L.InvestmentRatio	0.622*** (0.108)	0.616*** (0.107)	0.714*** (0.100)	0.612*** (0.104)
NetIncome	0.000*** (0.000)	0.000*** (0.000)	0.000 (0.000)	0.000*** (0.000)
FinancialCrisis2009	-1.181** (0.442)	-1.176** (0.448)	-1.391 (0.808)	-1.272** (0.517)
PC1_Aasian_Markets	-0.089* (0.051)	-0.095* (0.052)		
Asia_2016_2024	0.037 (0.133)			
VIX	0.009 (0.007)	0.008 (0.007)	0.007 (0.004)	0.003 (0.008)
Europe	-0.000 (0.006)	0.000 (0.006)	0.004 (0.011)	-0.001 (0.006)
Asia_NA_2016_2024		0.131 (0.172)		
NorthAmerica			0.004 (0.012)	
SP500			0.000 (0.000)	
Bovespa				0.000 (0.000)
MSCIEmergingMarkets				-0.001*** (0.000)
LATAM_dummy				-0.132* (0.064)
Constant	0.824** (0.377)	0.798** (0.361)	0.242 (0.563)	2.161*** (0.589)
Observations	275	275	167	275
R-squared	0.431	0.432	0.474	0.453
Number of firmid	19	19	14	19
Robust standard errors in parentheses				
*** p<0.01, ** p<0.05, * p<0.1				

*Correlation between the Bovespa Index and MSCI Emerging Markets was 0.43

The results from these 4 regressions offer insightful moves along the previous hypotheses. First, in (1), even though not significant, the sign of the coefficient on entities owing a share of their revenue to the Asian continent is positive, which could indicate that indeed these companies invest more. Similarly, when looking at companies with an interactive term between the period 2016 to 2024, the Asian region, and the North American region, that coefficient increases in size. Having an increasing Investment Ratio is further tested in (3), providing a case for regression (3) in which the same coefficient is estimated for North America, but this time the sign is still positive but lower and not significant, and even other

historically significant variables such as *Net Income* or the Financial Crisis lose their significance, which I believe could have also arisen by including the S&P500 Index and its exponential growth in recent years, despite it not being significant and close to 0. Finally, maybe the most significant result is that obtained in (4) with two significant variables that include the Index for Emerging Markets, with a negative sign which poses the question of knowing if the performance of less developed markets makes investment levels of CAC 40 firms less determinant than having a share of revenue in a more developed economy. The hypothesis that having a share of revenue in Latin America is significant at the 10% level, confirming the hypothesis from the earlier observations.

8 - A REGIONAL ANALYSIS OUTSIDE OF EUROPE

The Regional Analysis will again a two-step aim, to determine whether in different regions the link to French indicators such as the consumer sentiment is maintained, and depending on the region, which would be the strongest market driver of investment levels.

Asia

Table 13: Determinants of Investment - Asia Analysis: Consumer Sentiment			
VARIABLES	(1) Control Factors	(2) Consumer Sentiment	(3) Consumer Sentiment
L.InvestmentRatio	0.643** (0.212)	0.605** (0.212)	0.614** (0.204)
NetIncome	0.000*** (0.000)	0.000*** (0.000)	0.000*** (0.000)
FinancialCrisis2009	-0.509 (0.377)		
ratesXperiod_2010_2015	-0.050 (0.056)		
PC1_Aasian_Markets	-0.061 (0.068)		
VIX	0.013 (0.008)	0.004 (0.005)	-0.006 (0.005)
AsiaPacific	0.006 (0.012)	0.023 (0.017)	0.022 (0.016)
PC1_Aasian_Markets_2007_2015		-0.141 (0.099)	-0.318*** (0.078)
PC1_Aasian_Markets_2016_2024		0.017 (0.076)	-0.018 (0.081)
France_consumer_sentiment		-0.025* (0.012)	
consumer_sentiment_2007_2015			-0.008 (0.008)
consumer_sentiment_2016_2024			-0.036* (0.017)
Constant	0.430 (0.846)	-0.338 (1.176)	-0.149 (1.054)

Observations	144	144	144
R-squared	0.343	0.352	0.365
Number of firmid	10	10	10
Robust standard errors in parentheses			
*** p<0.01, ** p<0.05, * p<0.1			

The first regression testing for the control factors doesn't yield any significant results, which could be a result by noticing that firms who generate a high share of their revenue in Asia were not as dependent on the effects of the financial crisis in reducing their Capex investments, while the lack of significance from the VIX and the determined PC1. However, judging from (2) and (3) that in both cases exhibit a significant regressor from French consumer confidence or the PCA created for Asian markets and shown in Appendix 3, and taking into account the fact that the model's R-squared increasing suggests that the Asian markets performance as well as the consumer sentiment maintain a relationship with firms' investment decisions. Despite the coefficient being negative, that significance traces the relationship that exists.

However, knowing that firms such as Louis Vuitton or Kering are dependent on Japan because of their consumer basis, reported as if Japan was as important as France, we now wish to determine which of the Asian markets are the most determinant for the sampled firms, breaking away from PCA and rather considering individual markets characteristics, even if multicollinearity arises.

Table 14: Determinants of Investment - Asia Analysis: Markets					
VARIABLES	(1) Japan	(2) Asia Excluding Japan	(3) China	(4) India	(5) Japan & Luxury Sector
L.InvestmentRatio	0.649** (0.212)	0.604** (0.206)	0.632** (0.200)	0.640*** (0.185)	0.663*** (0.202)
NetIncome	0.000*** (0.000)	0.000*** (0.000)	0.000*** (0.000)	0.000* (0.000)	0.000** (0.000)
NIKKEI	-0.001 (0.002)				
VIX	0.018* (0.009)	0.011 (0.008)	0.009 (0.008)	0.024** (0.008)	0.019** (0.008)
AsiaPacific	0.005 (0.011)	0.026 (0.021)	0.018 (0.022)	0.006 (0.020)	0.011 (0.010)
MSCIAsiaexclJapan		-0.001* (0.001)			
ShanghaiComposite			-0.002*** (0.001)		
NIFTY				0.002 (0.002)	
LuxurySectorxNIKKEI					-0.003 (0.005)
Constant	0.432 (1.111)	0.692 (1.272)	1.193 (1.003)	-0.059 (1.286)	0.196 (0.744)
Observations	144	144	144	144	144
R-squared	0.337	0.351	0.366	0.334	0.341

Number of firmid	10	10	10	10	10
Robust standard errors in parentheses					
*** p<0.01, ** p<0.05, * p<0.1					

(1) is also interesting to look at depending on the regions: here we see that the firms concerned did not see a very large amount of CAPEX decrease due to the financial crisis, unlike every other case observed before.

In (2), I find it logical that the MSCI is relevant, although the sign is negative, and it is strange that in (1) the Nikkei is not significant. For Japan, we also wanted to look at the firms in our data who are highly dependent on Japan, like Kering and LVMH who report the numbers for Japan, and Hermès is also included in that dummy variable, despite it not reporting the numbers (but its share of revenue in Asia in that year was 50%, so we imagine that it has a significant share in Japan).

Where this is also very significant is in seeing the significance of the coefficient of China, which we know is an important client of the CAC 40 firms, in the luxury sector but not only. Finally, we see that the coefficient for the NIFTY is not significant, which could either indicate a lack of exposure compared to the other markets or could be a misspecification like it is probably the case with Japan.

North America

Table 15: Determinants of Investment - NorthAmerica Analysis			
VARIABLES	(1) Control Factors	(2) Consumer Sentiment	(3) Consumer Sentiment
L.InvestmentRatio	0.722*** (0.112)	0.694*** (0.120)	0.693*** (0.124)
NetIncome	0.000** (0.000)	0.000** (0.000)	0.000** (0.000)
FinancialCrisis2009	-0.710* (0.350)	-0.918* (0.457)	-1.001* (0.485)
SP500_2007_2015		0.000 (0.000)	-0.000 (0.000)
SP500_2016_2024		0.000 (0.000)	0.000 (0.000)
France_consumer_sentiment		-0.034** (0.015)	
VIX	0.018*** (0.006)	0.005 (0.007)	-0.000 (0.008)
NorthAmerica	0.011 (0.011)	0.009 (0.010)	0.016 (0.011)
ratesXperiod_2010_2015	0.138 (0.078)		
SP500	0.000 (0.000)		
consumer_sentiment_2007_2015			-0.064** (0.025)
consumer_sentiment_2016_2024			-0.019 (0.011)

Constant	-0.157 (0.633)	-0.015 (0.598)	-0.155 (0.608)
Observations	199	199	199
R-squared	0.472	0.484	0.494
Number of firmid	14	14	14
Robust standard errors in parentheses			
*** p<0.01, ** p<0.05, * p<0.1			

It is important to include the Financial Crisis, I believe, because it is where the crisis erupted; however, in the controls, seeing the significance of the VIX variable, it must be absorbing the effects of the other variables.

We are happy to see that there is a 10% significance for North America, which is higher than the result obtained for Asia (10%), which could indicate that because, on average, companies have higher shares of revenue in North America, there is more of an interconnection between the two economies and the specified firms.

Lastly, we see that there is significance at the 10% level in the first period of our dataset, as if it were determined by the crisis.

LATAM

We will not be removing companies such as ENGIE, Carrefour, Danone, which have performed poorly because it lies at the center of one of our hypotheses for the region, which might be that it is at the root of weaker performance for this dataset.

Table 4: Determinants of Investment - Latin America Analysis			
VARIABLES	(1) Control Factors	(2) Consumer Sentiment	(3) Consumer Sentiment
L.InvestmentRatio	0.332** (0.106)	0.332** (0.095)	0.311** (0.093)
NetIncome	0.000*** (0.000)	0.000*** (0.000)	0.000*** (0.000)
FinancialCrisis2009	0.459 (0.384)	0.458 (0.358)	0.256 (0.389)
ratesXperiod_2010_2015	0.227** (0.086)	0.227** (0.073)	0.153 (0.103)
Bovespa	-0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)
MSCIEmergingMarkets	-0.002** (0.000)	-0.002* (0.001)	-0.001* (0.001)
consumer_sentiment_2007_2015			-0.010 (0.022)
consumer_sentiment_2016_2024			0.004 (0.019)
VIX	-0.020** (0.007)	-0.020* (0.008)	-0.023** (0.009)
LatinAmerica	0.003	0.003	0.002

	(0.003)	(0.002)	(0.002)
France_consumer_sentiment		-0.001	
		(0.019)	
Constant	4.077***	4.067***	4.058***
	(0.349)	(0.578)	(0.562)
Observations	87	87	87
R-squared	0.467	0.468	0.470
Number of firmid	6	6	6
Robust standard errors in parentheses			
*** p<0.01, ** p<0.05, * p<0.1			

These are very interesting results, starting with the control factors: the firms are not really affected by the Financial Crisis (or we see weird patterns for ENGIE). The variable is very sensitive to interest rates compared to others. Bovespa does not seem to be highly significant, which could indicate that Brazil is not the main customer of the CAC 40 selected firms. French consumer confidence is not significant here in any of the two cases, and it is the first time that this happens.

9 – LIMITATIONS AND FURTHER RESEARCH IDEAS

Statistical Methodologies

It would have been interesting to apply regression analysis to stock prices or levels of investing while accounting for the dependence on lagged variables by using a preliminary check based on the ARMA methodology developed by George E. P. Box and Gwilym Jenkins in 1970. Differencing or applying the logarithm to ensure stationarity for each company's Stock Price and CAPEX and identifying the level of autoregressive and moving average terms, would have been a necessary step in determining whether the variable is mainly determined by its lagged value, which could be a reasonable assumption for investing, or rather by a confluence of factors for stock prices. However, having only 17 observations makes this approach weaker, but were there to be a relationship, it would still be important to attempt to account for it.

Depending on the results of such an ARMA model, for a specific entity or for groups of entities, the inclusion of lagged terms could lead to correlation between error terms and observations across time, because past values embed past shocks, which might lead to inconsistent fixed effects estimators, also known as Nickell bias. Thus, when analyzing regression output using lagged variables, it would have been necessary to test the autocorrelation between error terms to ensure that inference across periods remained valid.

Another method that would have been useful is the Arellano Bond methodology, designed to address correlation between terms through first differencing. However, this research faces a situation with few entities, 22, which would still work for the Arellano Bond approach, but also only 17 time periods, which, just like for the ARMA framework, reduces its effectiveness. Therefore, in this research, we will assume that our error terms are exogenous.

Inclusion of certain control variables

As discussed in the literature, multinational firms shift capital or profit generation abroad because of tax advantages. Thus, one of the variables missing to capture part of this international activity is an

indicator of the tax levels in each country where the firms operate, or a measurement of the benefits gained by the studied firms.

Support by the State, either through an explicit reported share acquired by the State or through a dummy variable indicating whether the French State has acquired a position, is also important. Airbus naturally comes to mind, because it is supported not only by the French government but also by a European coalition to make it an international champion, which undoubtedly influences investors' views and decision making. At the moment, for example, due to its poor performance, ArcelorMittal's possible nationalization is at the core of the French political debate, which could impact its performance and investor sentiment.

To complement this research, I believe that extensive qualitative analysis is required for all the companies studied. This includes understanding their culture, the demands from shareholders and the objectives of management, changes in the regulatory environment, or even whether they have been affected by scandals throughout the years, as is sometimes the case for CAC 40 companies. All these elements could explain a higher exposure to different regions or to movements in asset prices and decision making.

Precision of the data

We do not always have the exact share of revenue generated in France. Often Europe is the basis for reporting by these firms, which has made it difficult to quantify the exposure to the French economy. Since Europe includes France, it becomes harder to know whether the share of revenue generated specifically in the country is minimal or, on the contrary, substantial enough to make the firm dependent on the economic and political state of the nation.

Despite more than 400 data points, the data remains a limitation for the CAC 40 companies, but also for French, European, and international indicators. Indeed, because of unbalanced data for the share of revenue generated abroad, the sample went from 36 companies, excluding banks as the literature suggested, to 22 companies, meaning that sectors as important as energy or industry are missing from the analysis. Companies such as TotalEnergies, Stellantis, or Thales, which represent a large share of the Index, are therefore not included.

This limitation also applies to French indicators. I believe that another significant improvement would have been the inclusion of additional indicators of the state of the French economy, including social indicators such as consumer sentiment or population happiness, as well as more specific indicators for different industries. I would have liked to include smaller companies in the same way as I presented the specifics of CAC 40 firms, meaning their Stock Price, CAPEX, Revenue Growth, and so on, whether listed or not. These firms might have generated a larger portion of their revenues in France and would not benefit from the same level of exposure to foreign investors. The difficulty with the CAC Small is that it still contains some of the largest companies in the country, which have access to international markets both commercially and financially.

Expanding the research to years prior to 2007, if data were available, or using quarterly or monthly data instead of yearly data, would also have enabled stronger statistical results. With more time observations, methods such as ARMA and GARCH models could have been used more effectively to capture links over time and volatility across international markets. Furthermore, the Arellano Bond methodology would have been entirely adequate here with more time periods and a relatively small number of entities to ensure that fixed effects were not correlated across time.

It might seem paradoxical that I am arguing that the French economy is not representative of other economies, and that the same might be true for other countries, while at the same time using the main stock indices of those countries as proxies for their economic state. In this sense, an interesting extension would be to conduct the same analysis for other countries to allow for meaningful comparisons and to determine what benchmark demonstrates whether a stock index is representative of an entire

economy. That could be done by taking into account each country's and each index's particularities. For example, the S&P 500 in the United States owes an important part of its growth in recent years to technology companies, unlike France. Studying these differences could reveal common indicators across countries that show stronger links with country specific factors, particularly during periods of national crisis.

In the same way and given the strong exposure of the CAC 40 to global luxury companies, it would be interesting to look specifically at the luxury sector to see whether the patterns identified here are stronger, weaker, or entirely different.

Finally, having collected this data, many other applications could be explored, such as equity research including predictions for future stock prices, or an event study based on the period covered, despite the difficulty of applying such methods with yearly data and the challenge of supporting the parallel trends assumption.

10 - CONCLUSION

Therefore, in a challenging international and macroeconomic environment, this study provides insights as to how well the CAC 40 and the French economy are connected.

It is very interesting to note that even when accounting for many factors, the CAC 40 seemed to remain relevant, and it was clearly observed that there were changes in the signs, coefficients, and significance when looking at different sectors and regions. This provides the foundation for further research that could be done in this field, as described in the limitations section and the further ideas, with the objective of including more statistical methods, more sectors, building the robustness of the data, adding variables, and in the long run applying the same methodology to other markets as well as tracking the results through time.

11 - APPENDIX

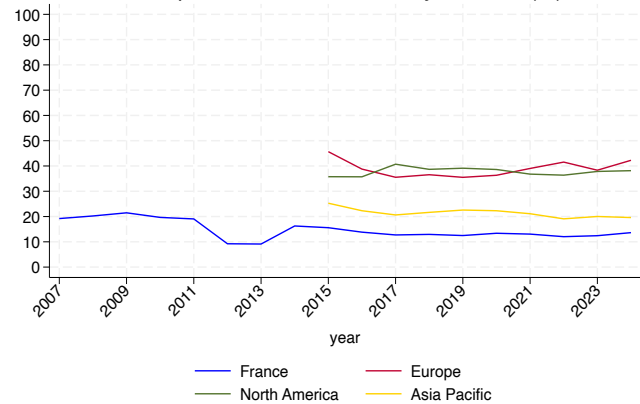
Appendix 1: Companies' Share of Revenue by Location from 2007 to 2024 (as reported), Share of Revenue by Location in 2024 (as reported), Stock Price: 2007-2024, Capital Expenditure: 2007-2024

Notes concerning the following graphs:

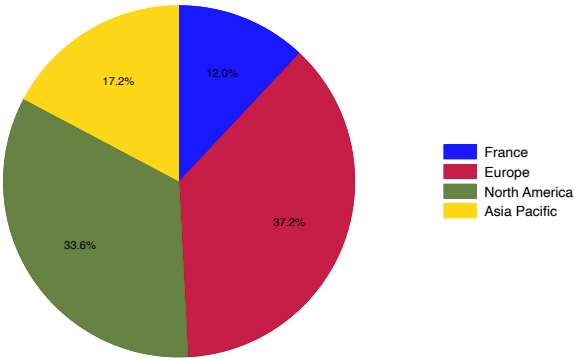
1. *France* is included in the variable *Europe* when both are reported, *Japan* as well when the country is reported together with *Asia Pacific*
2. If the sum of percentages doesn't reach 100% for a given year, or more importantly in 2024, it could indicate that a variable share for *OtherCountries* has not been calculated if it is below 100%, or on the contrary the variable has been counted twice if the sum goes past 100%.
3. When not mentioned as the title of the graph, *Last Price* or *Stock Price* is in US dollars, and *Absolute Capex* or *Capex* is in Millions of US dollars.
4. If the share of Revenue generated in the *United States* was reported, but not that of *North America*, *North America* is set equal to the level of *United States*, as a proxy.

Air Liquide

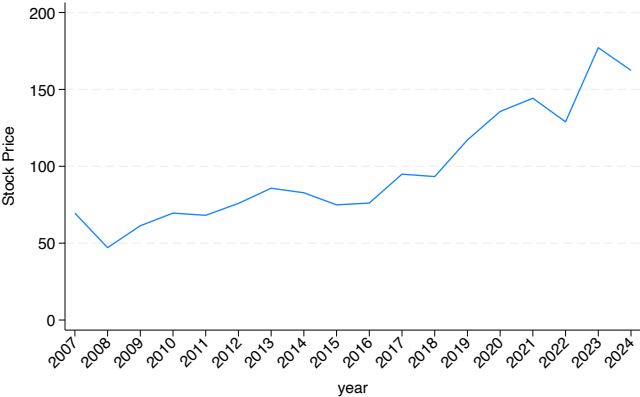
AirLiquide Share of Revenue by Location (%)



AirLiquide Share of Revenue by Location in 2024 (%)



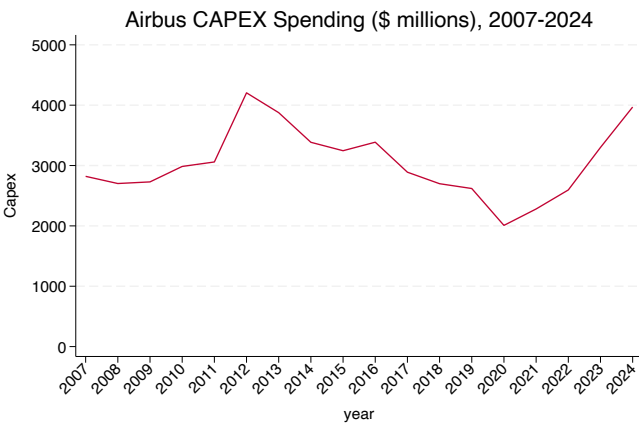
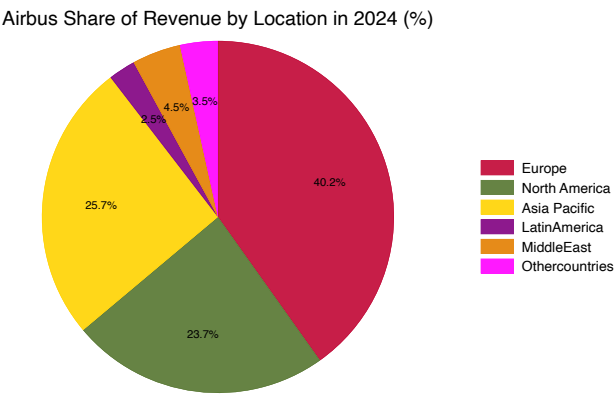
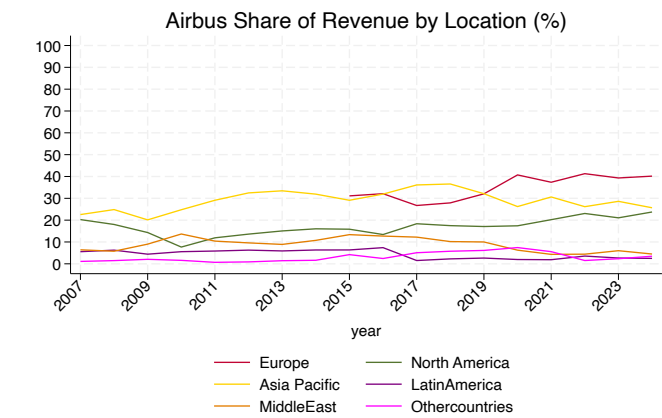
Air Liquide Stock Price (\$), 2007-2024



Air Liquide CAPEX Spending (\$ millions), 2007-2024

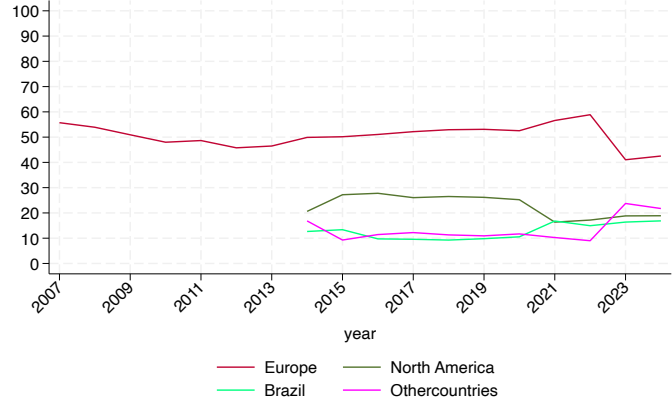


Airbus

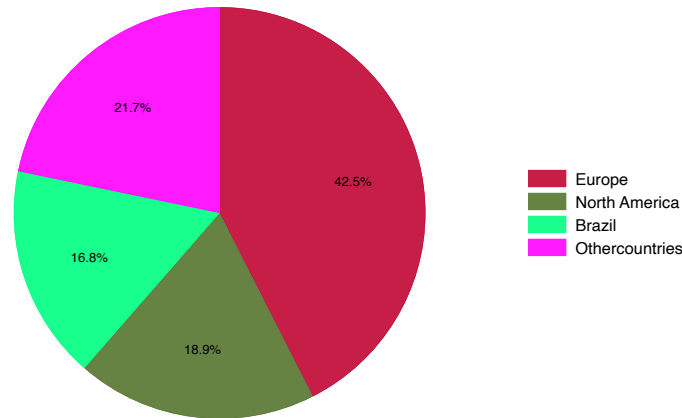


ArcelorMittal

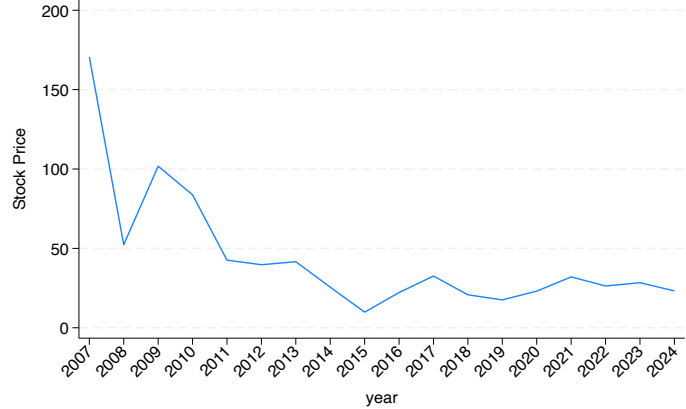
ArcelorMittal Share of Revenue by Location (%)



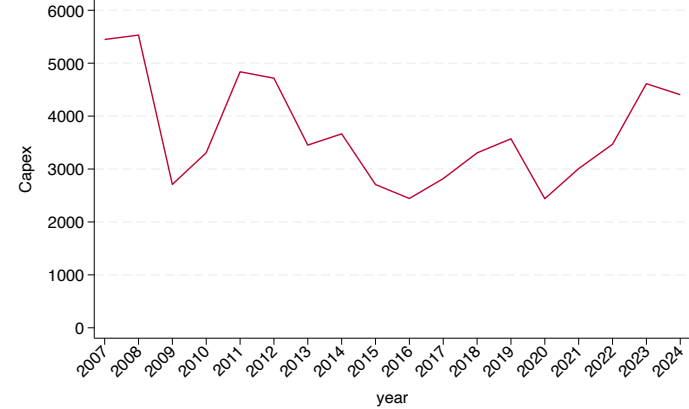
ArcelorMittal Share of Revenue by Location in 2024 (%)



ArcelorMittal Stock Price (\$), 2007-2024

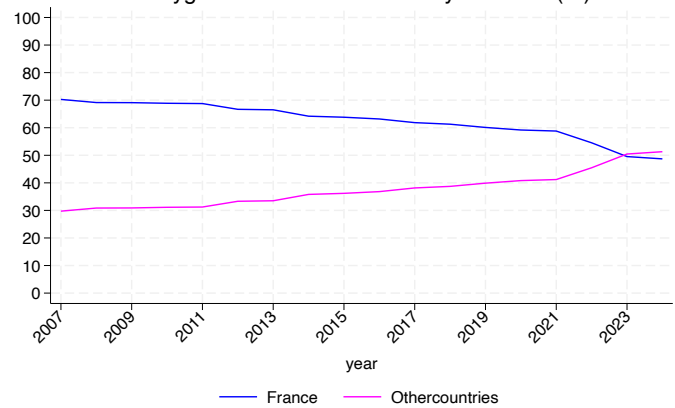


ArcelorMittal CAPEX Spending (\$ millions), 2007-2024

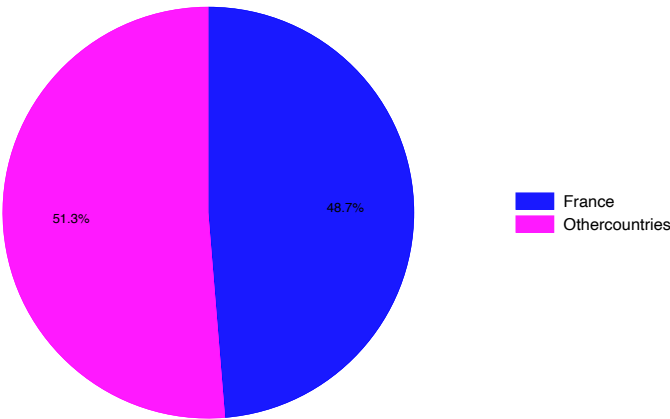


Bouygues

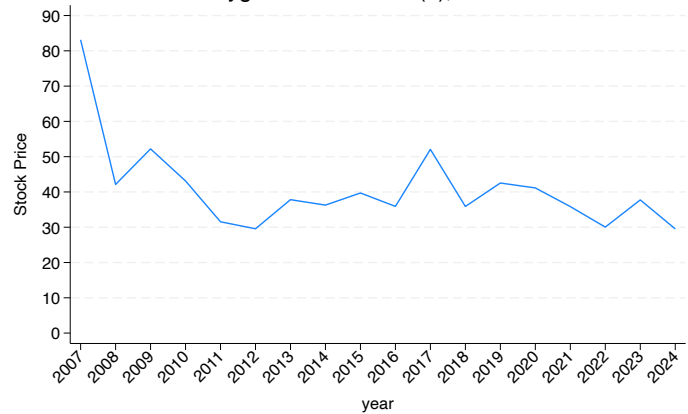
Bouygues Share of Revenue by Location (%)



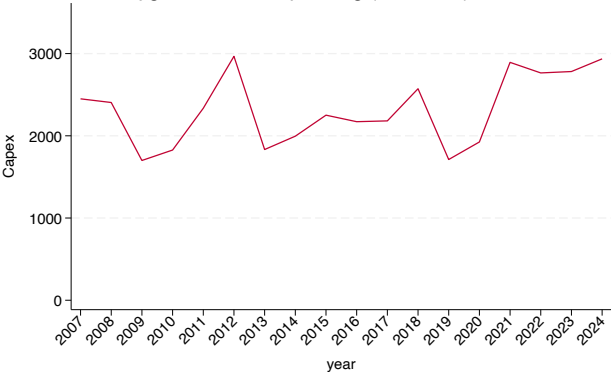
Bouygues Share of Revenue by Location in 2024 (%)



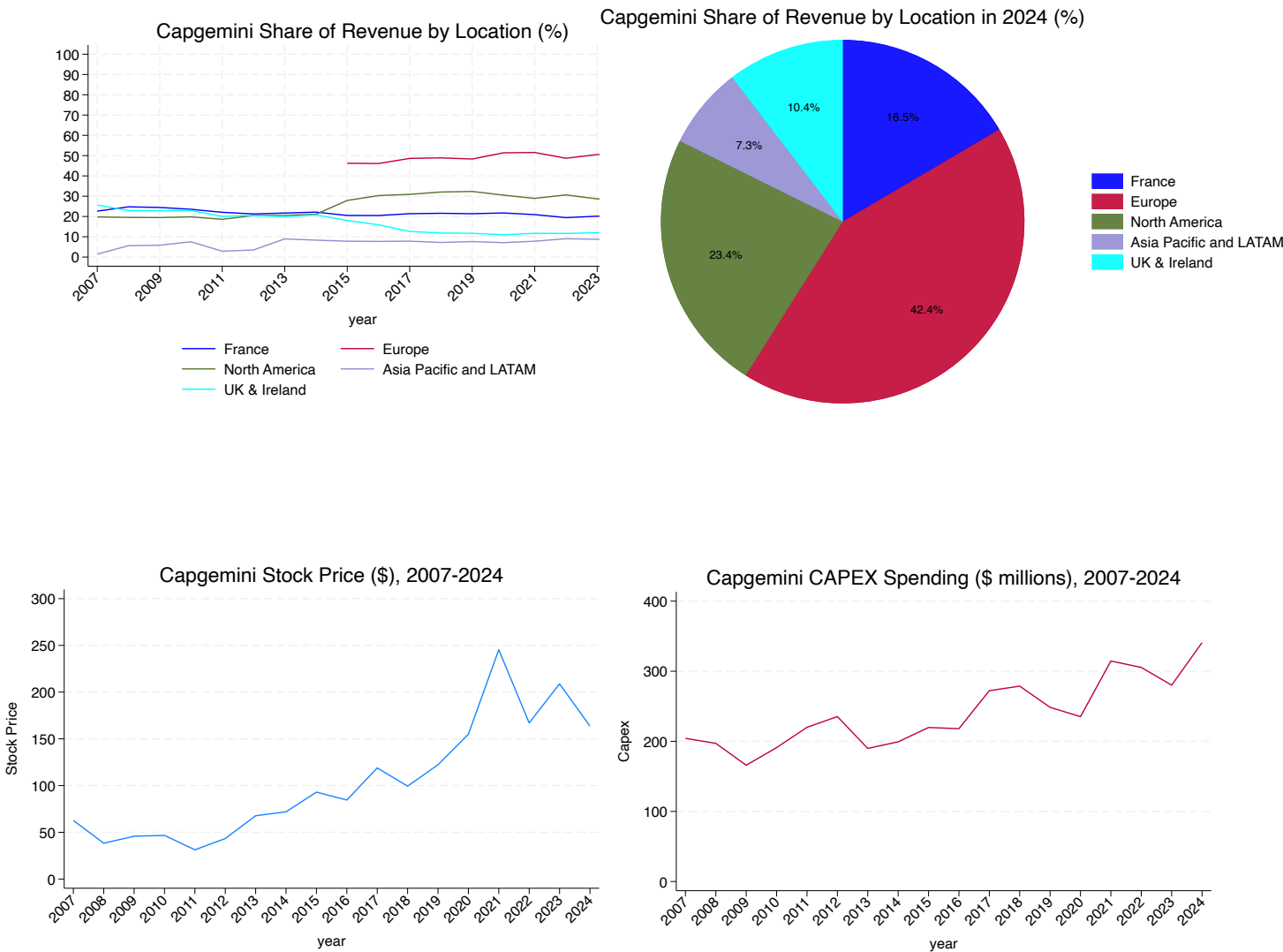
Bouygues Stock Price (\$), 2007-2024



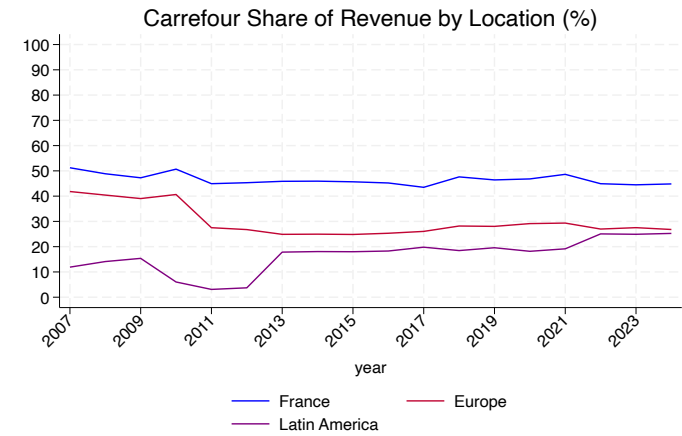
Bouygues CAPEX Spending (\$ millions), 2007-2024



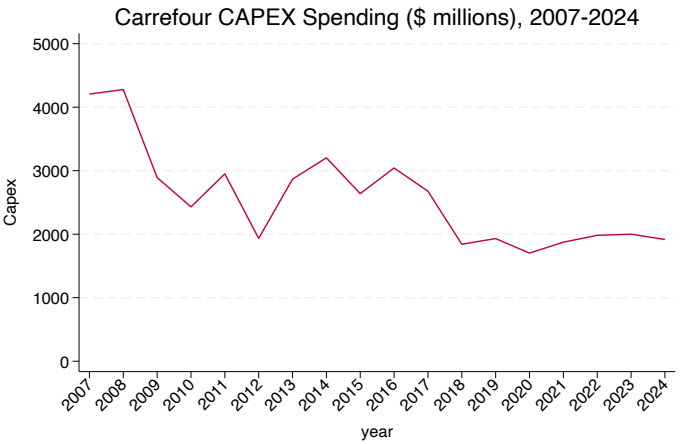
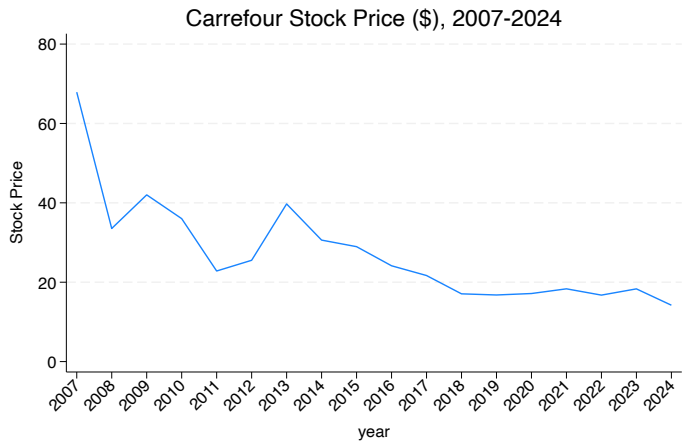
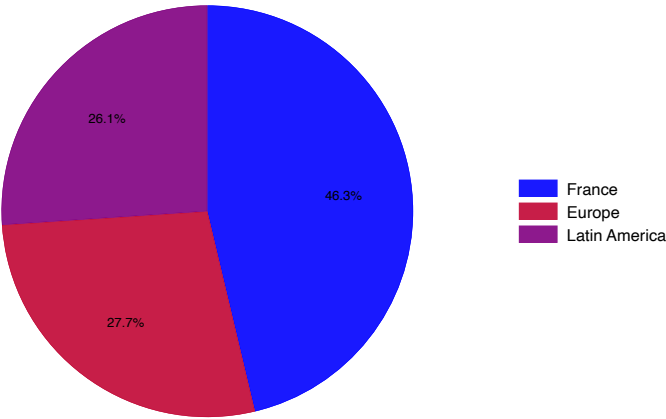
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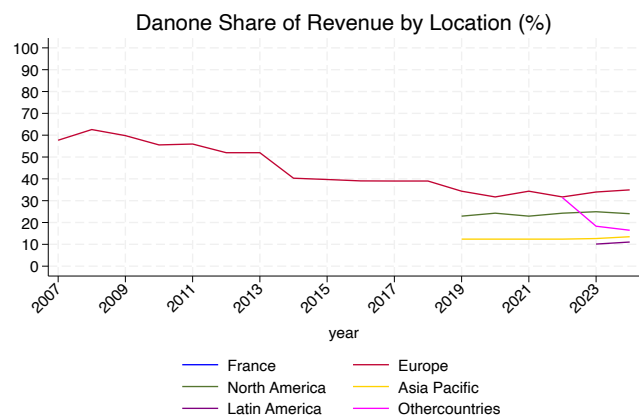
Carrefour



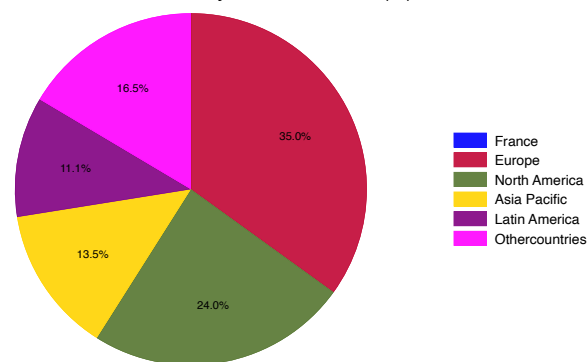
Carrefour Share of Revenue by Location in 2024 (%)



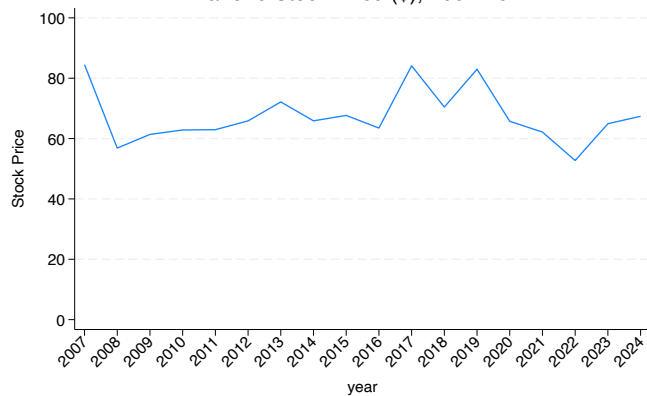
Danone



Danone Share of Revenue by Location in 2024 (%)



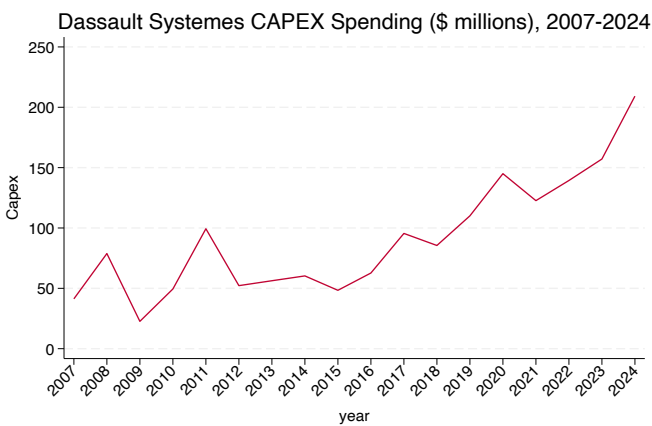
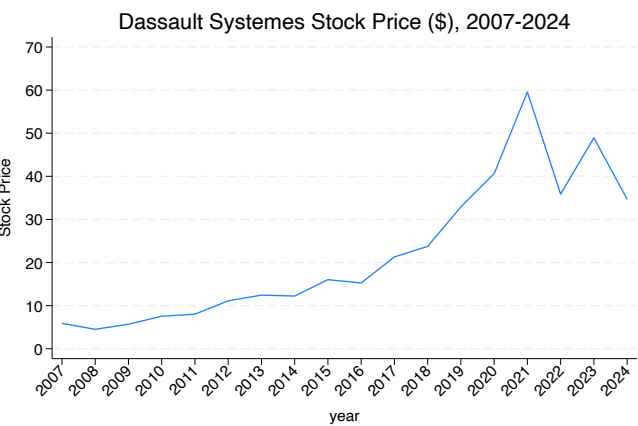
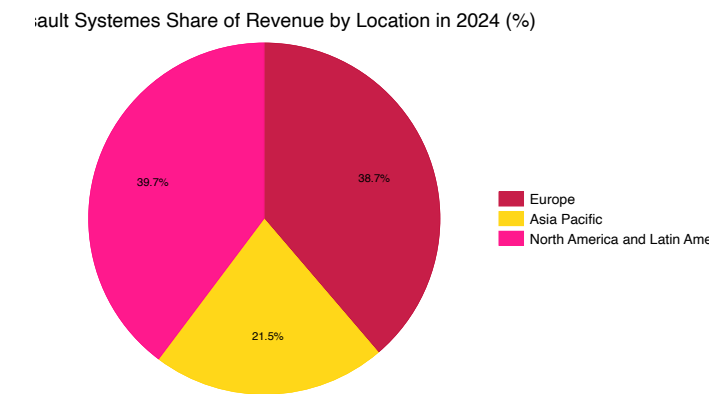
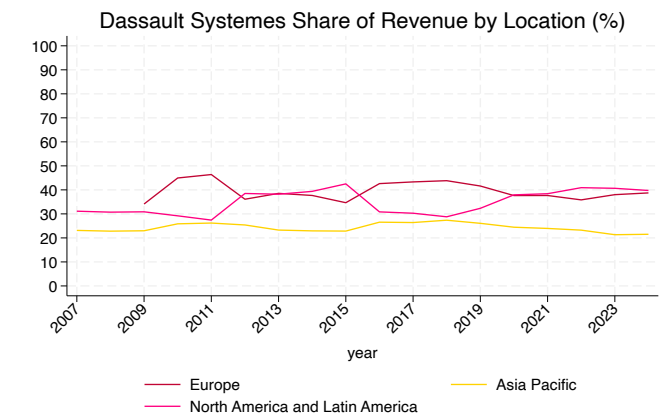
Danone Stock Price (\$), 2007-2024



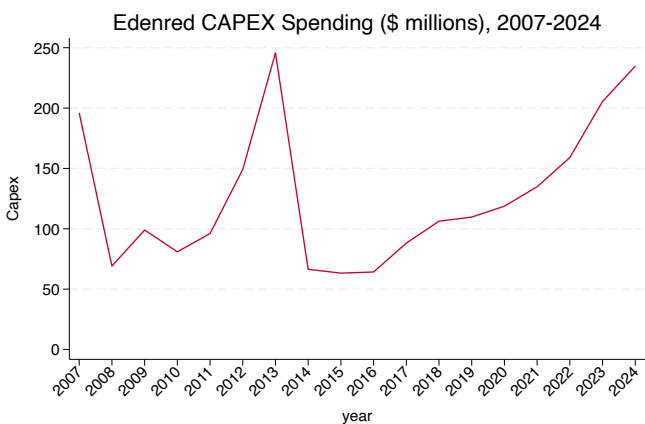
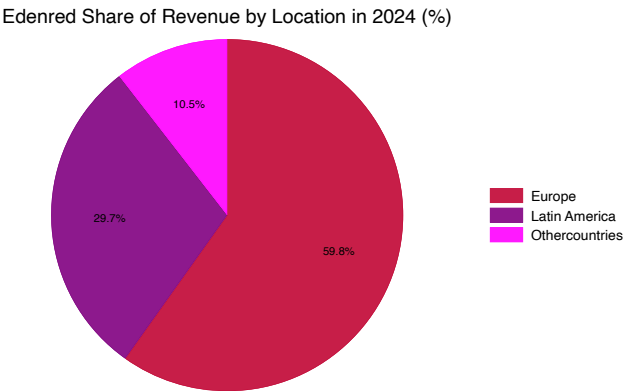
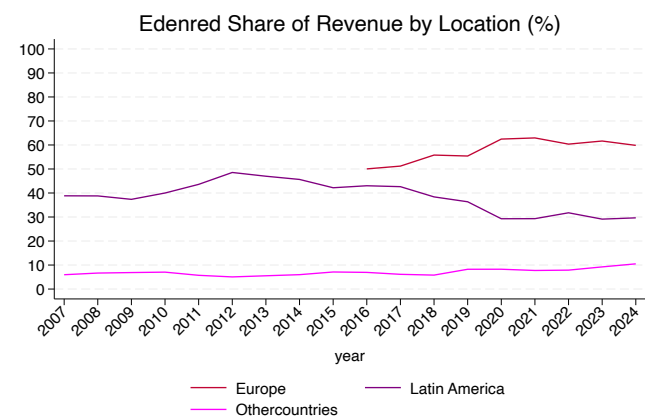
Danone CAPEX Spending (\$ millions), 2007-2024



Dassault Systemes

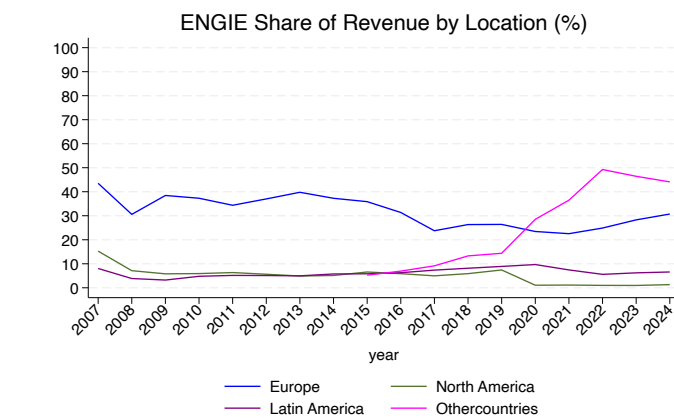


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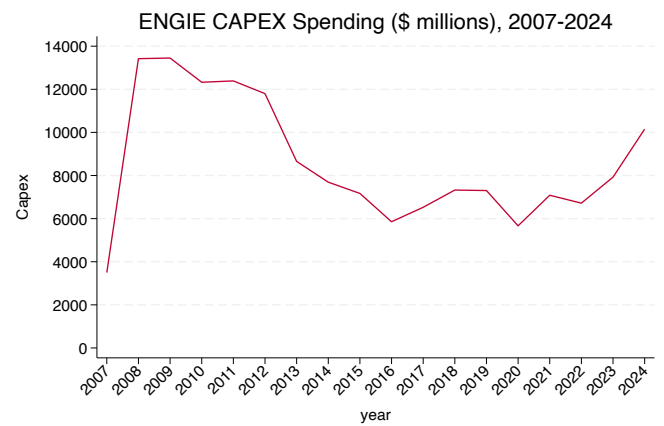
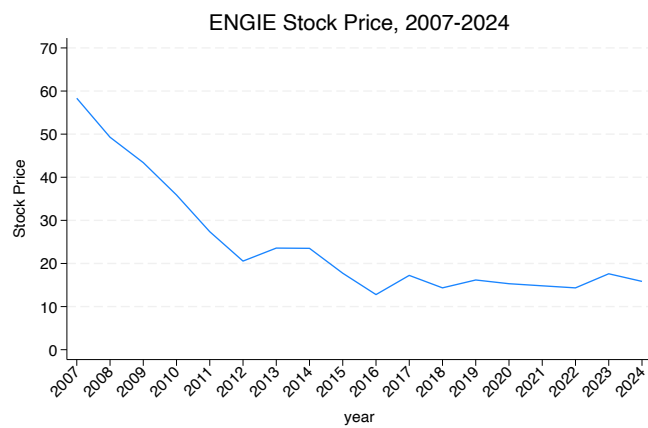
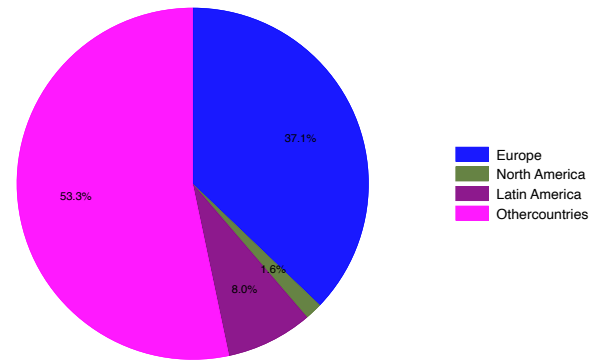


ENGIE

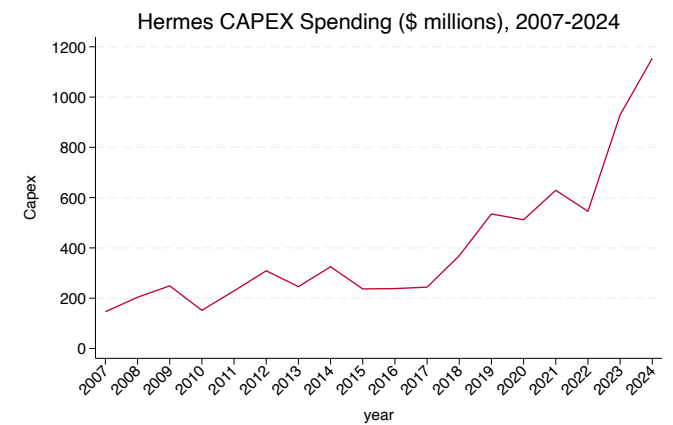
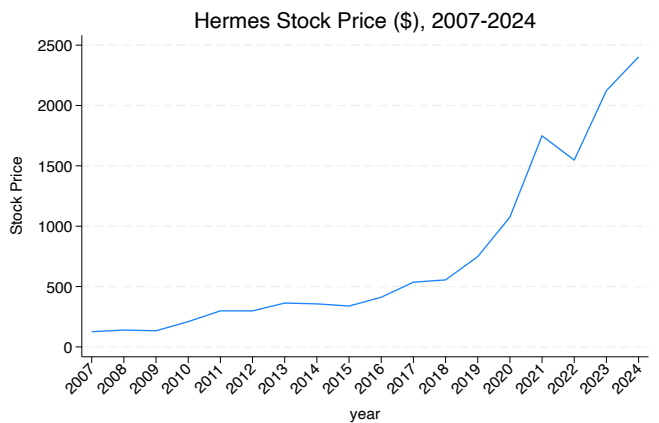
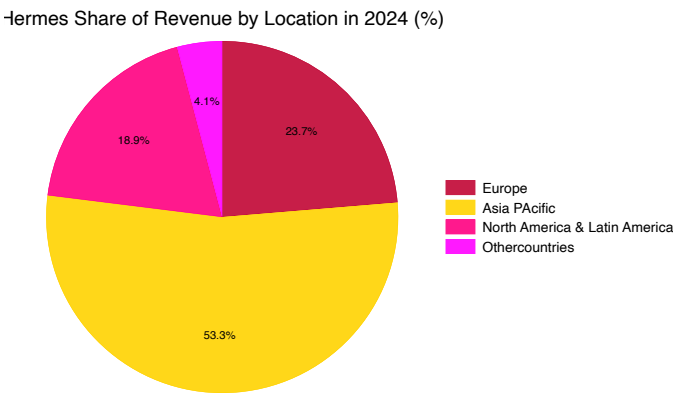
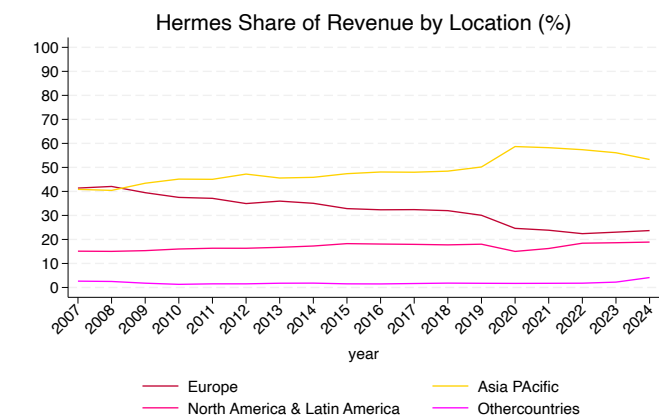
There is a mistake here, Europe is actually France



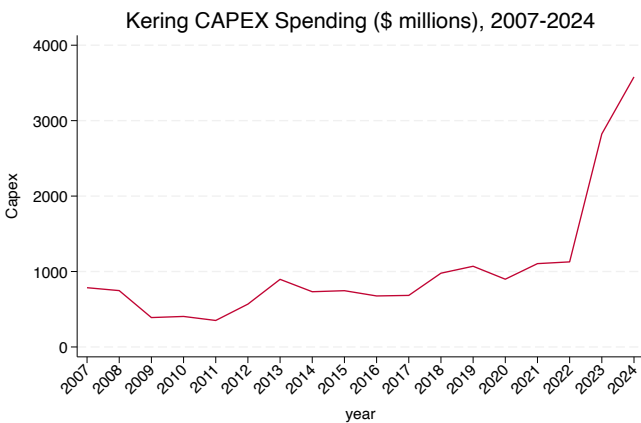
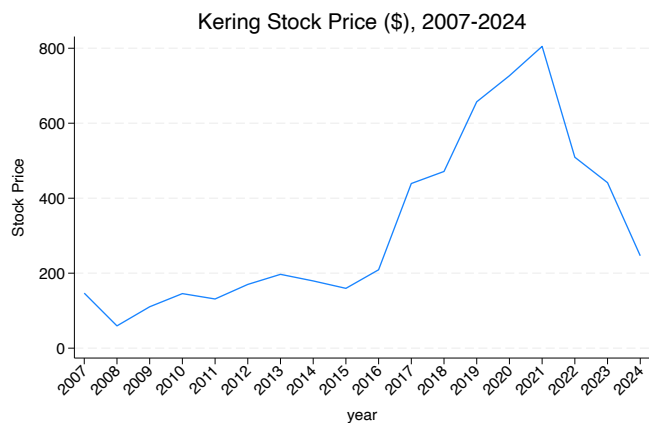
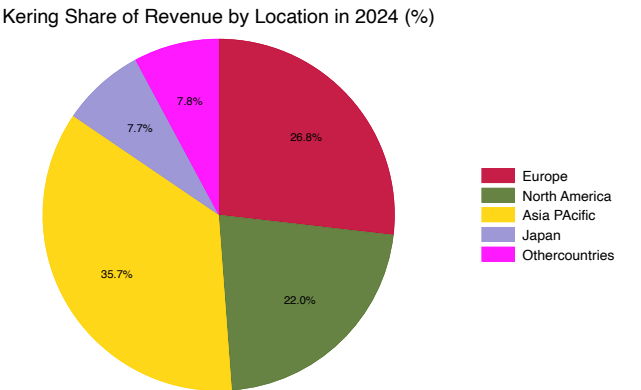
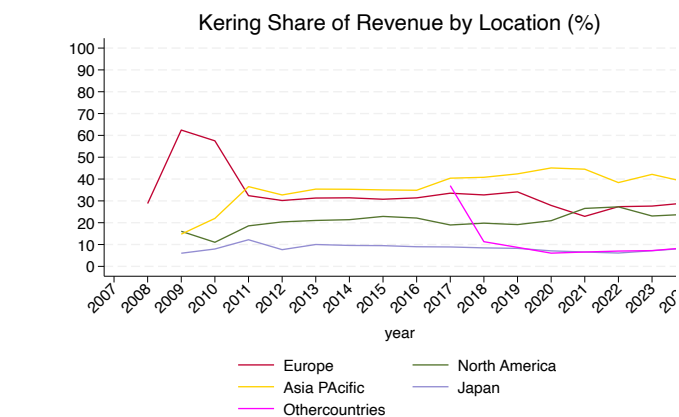
ENGIE Share of Revenue by Location in 2024 (%)



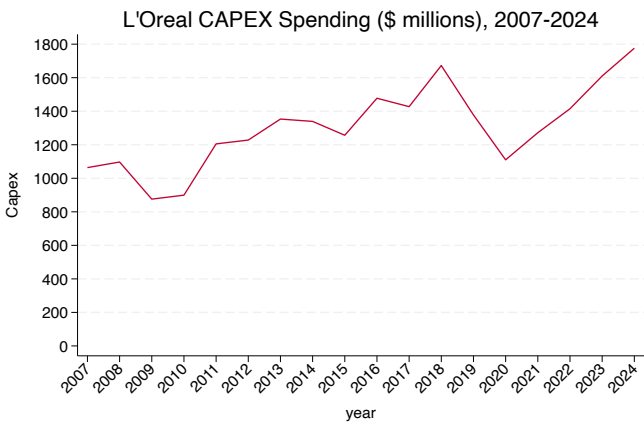
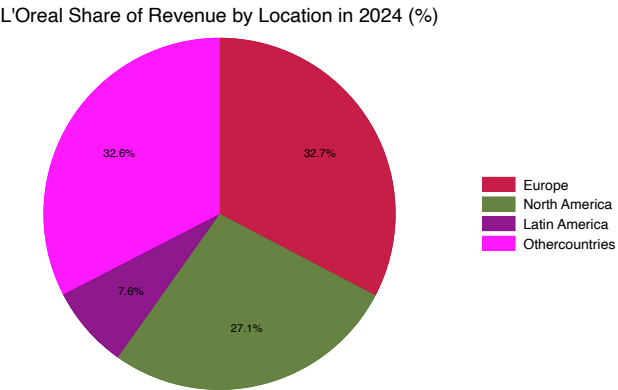
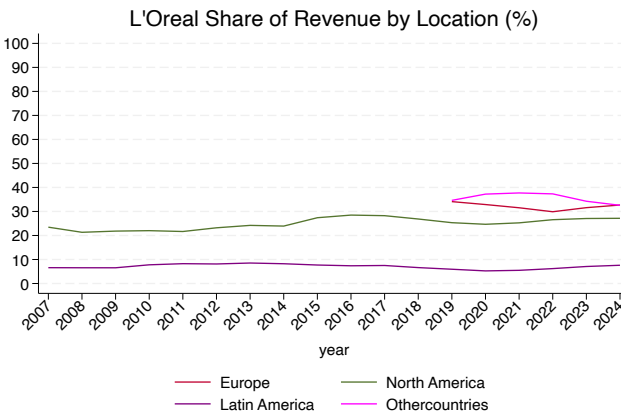
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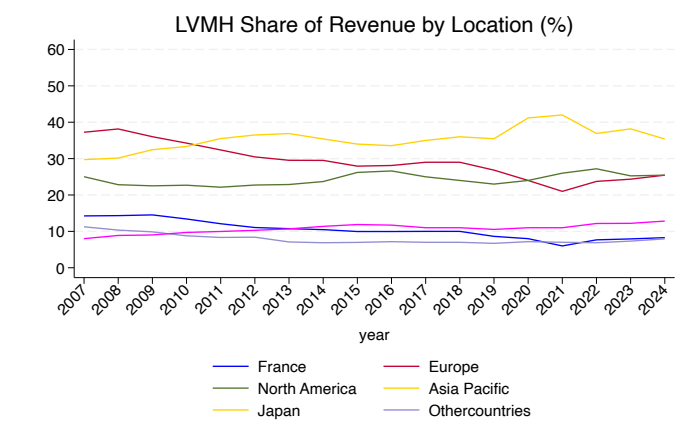
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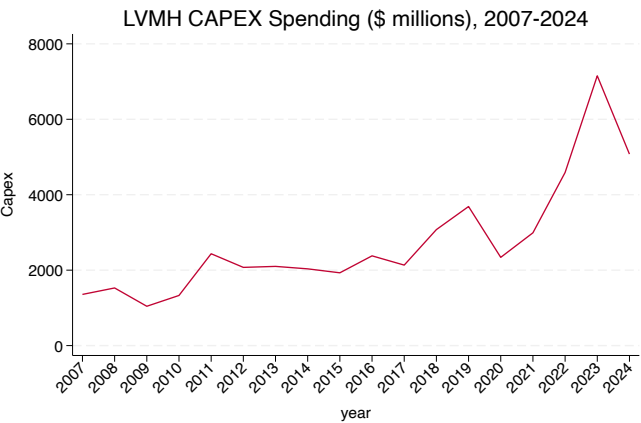
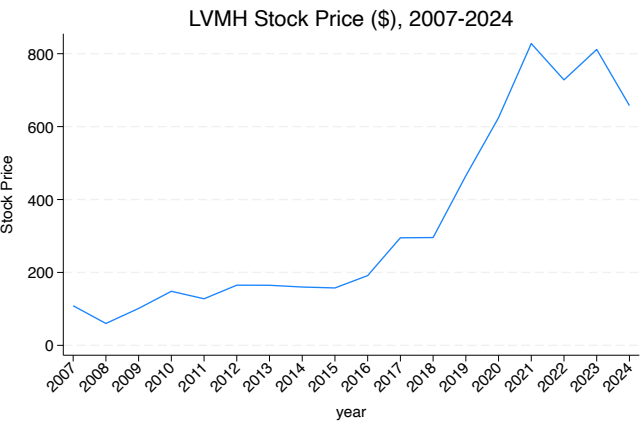
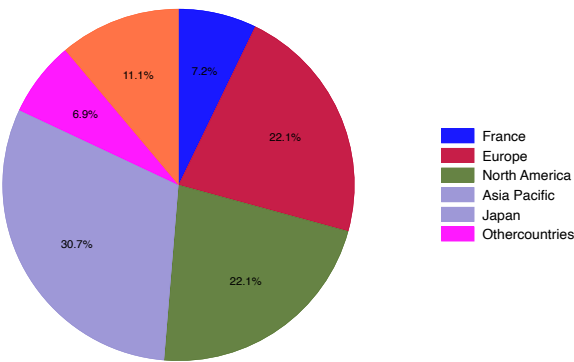
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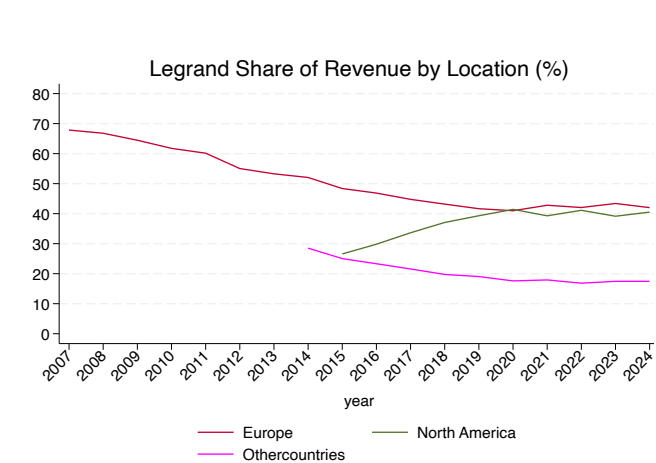
LVMH



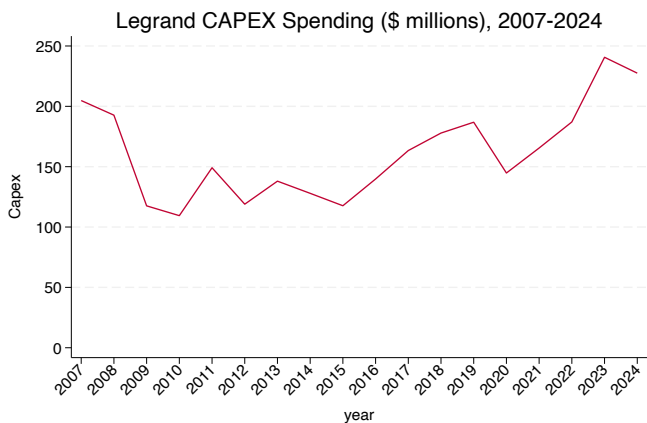
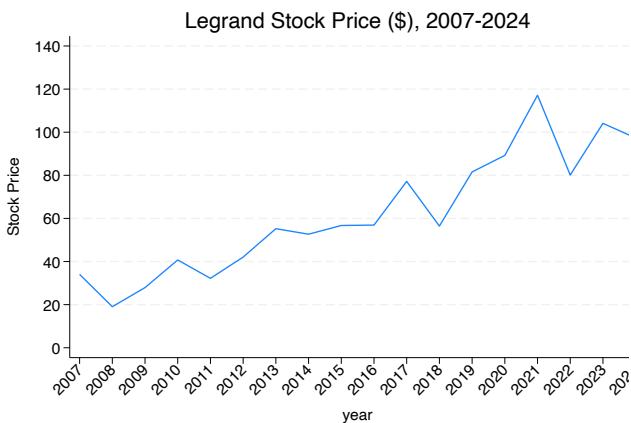
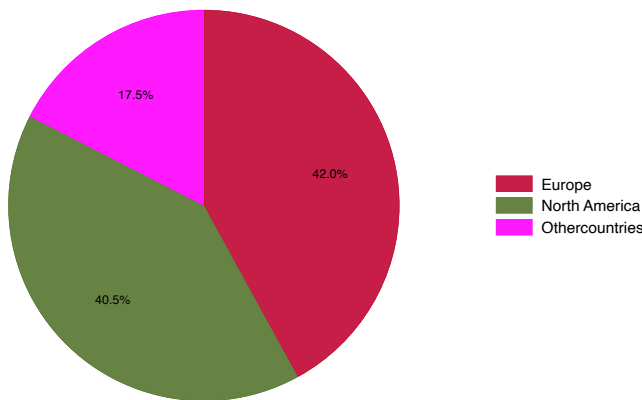
LVMH Share of Revenue by Location in 2024 (%)



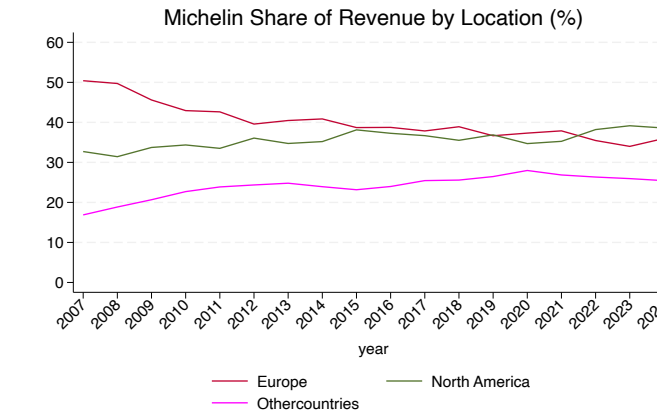
Legrand



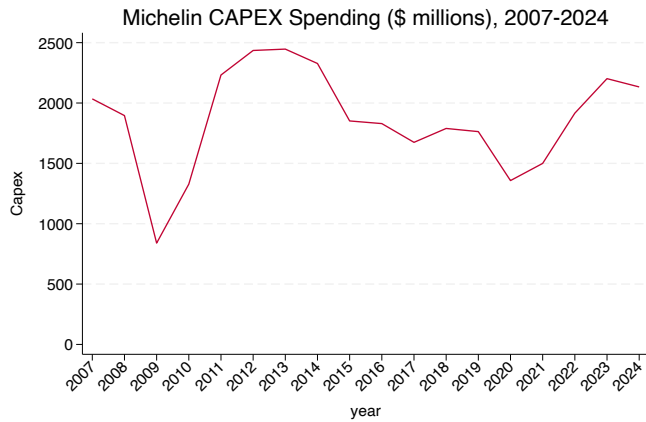
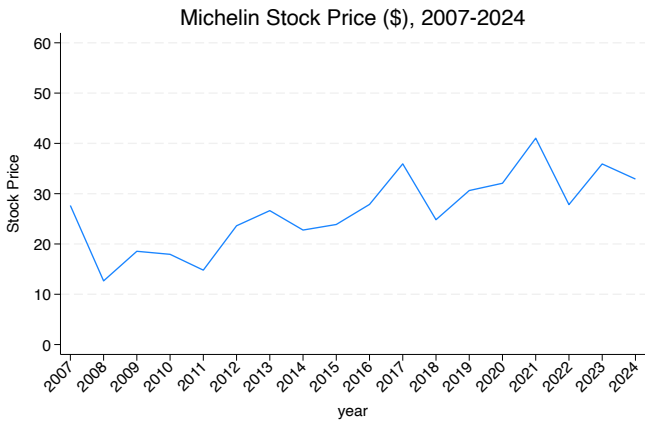
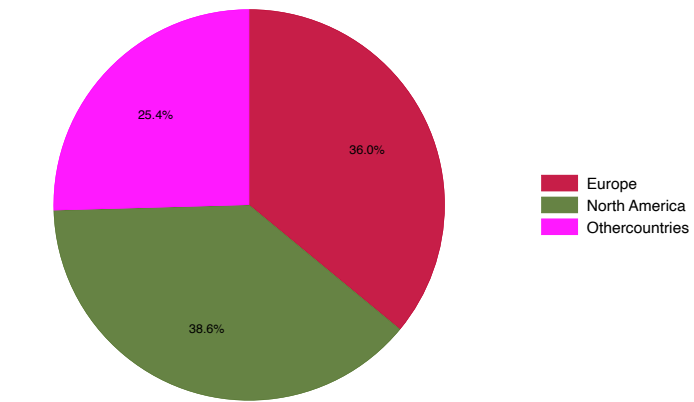
Legrand Share of Revenue by Location in 2024 (%)



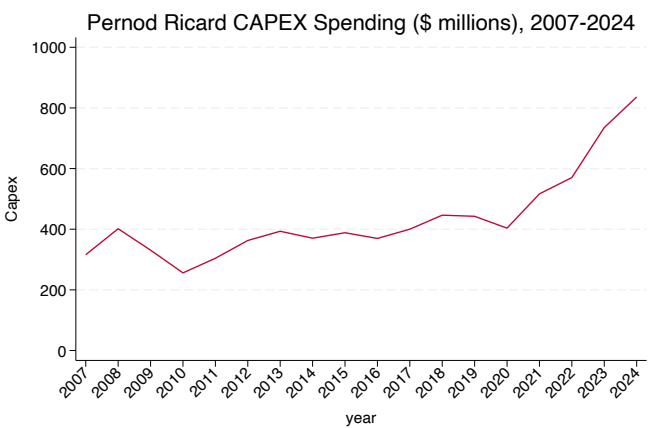
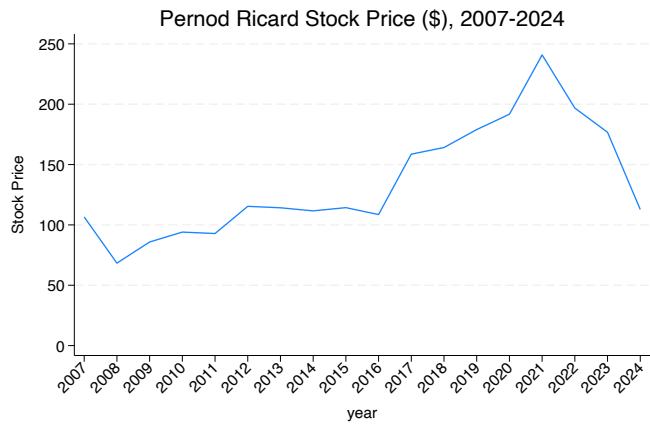
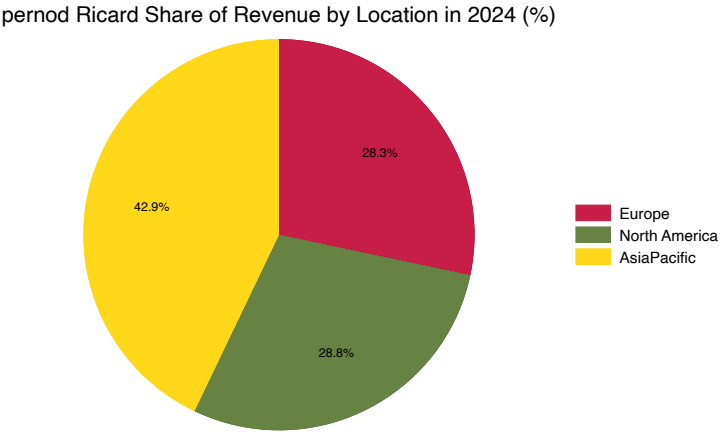
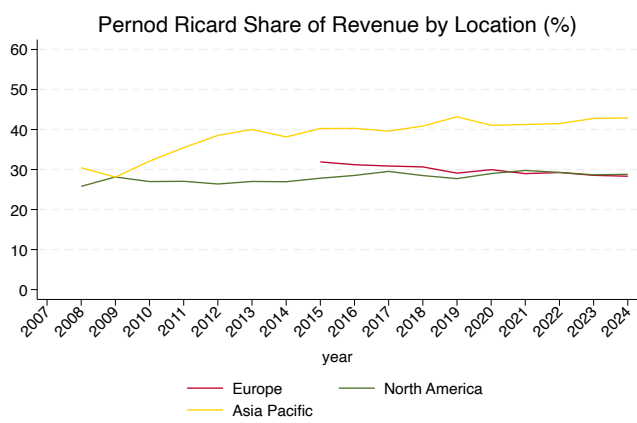
Michelin



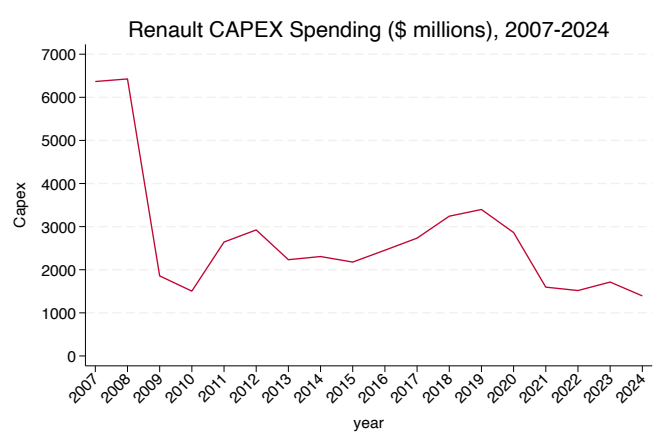
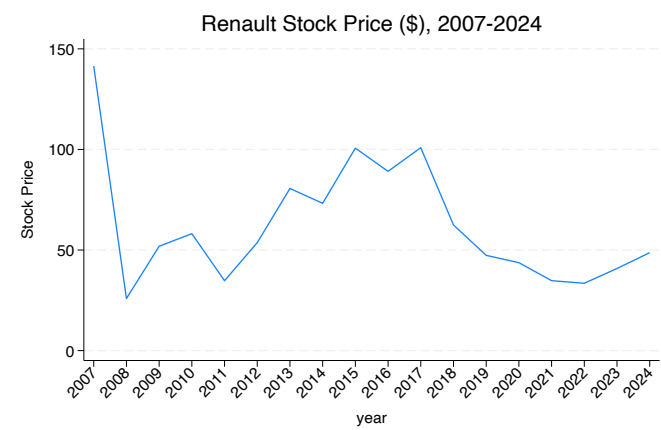
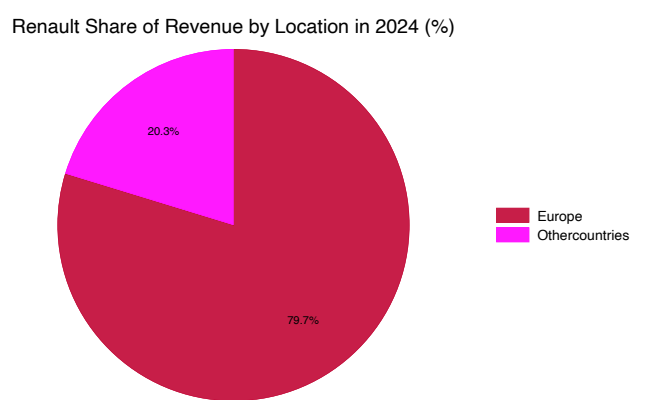
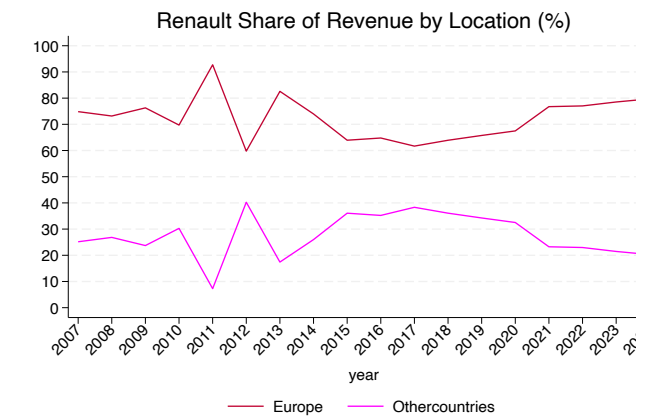
Michelin Share of Revenue by Location in 2024 (%)



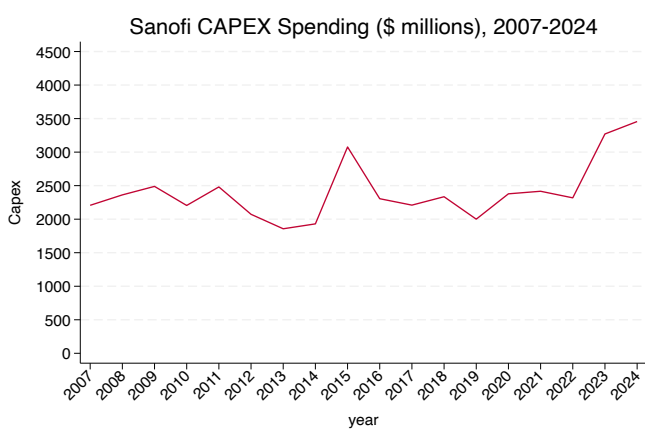
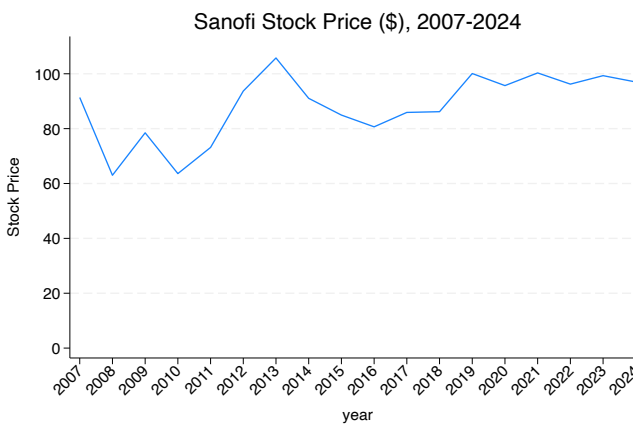
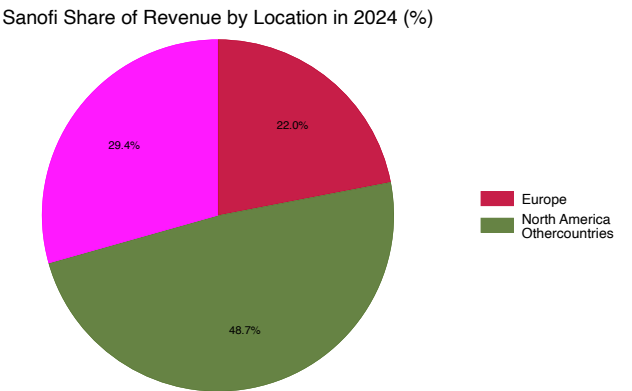
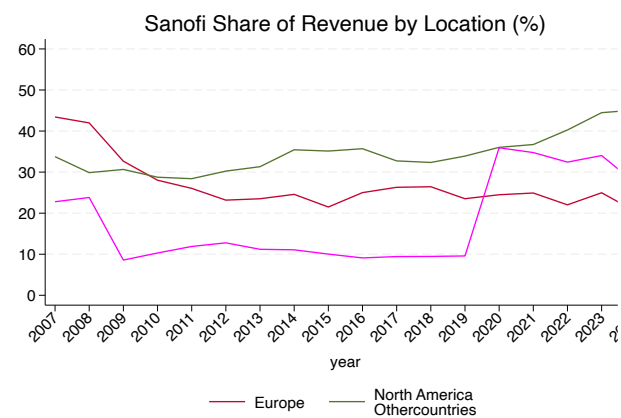
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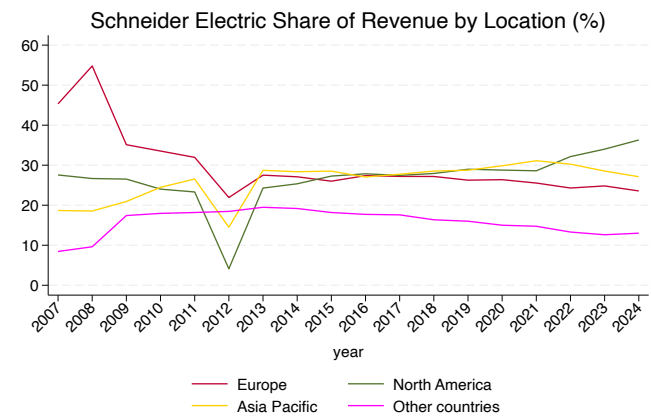
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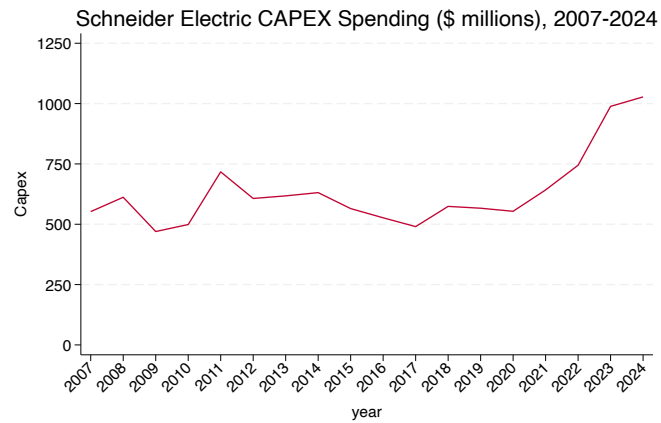
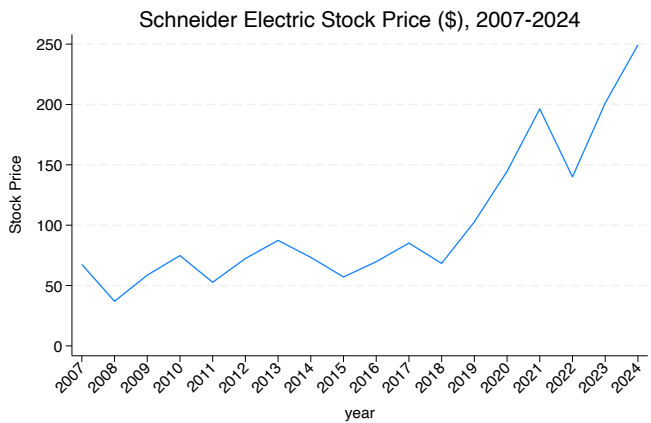
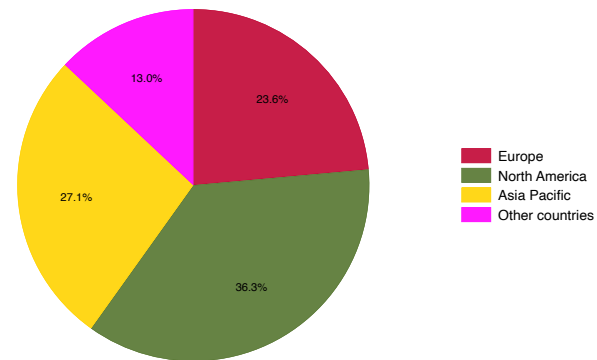
Sanofi



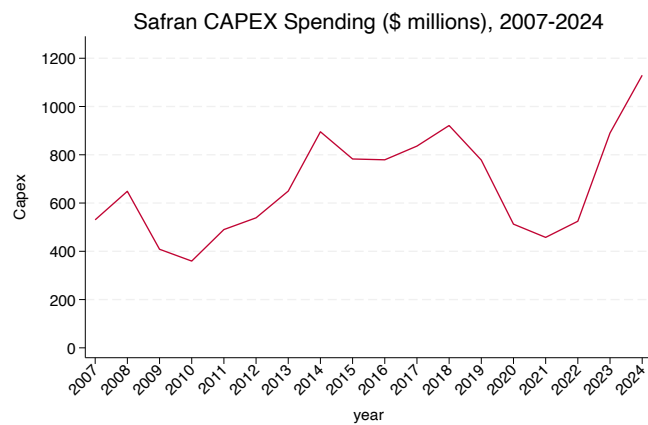
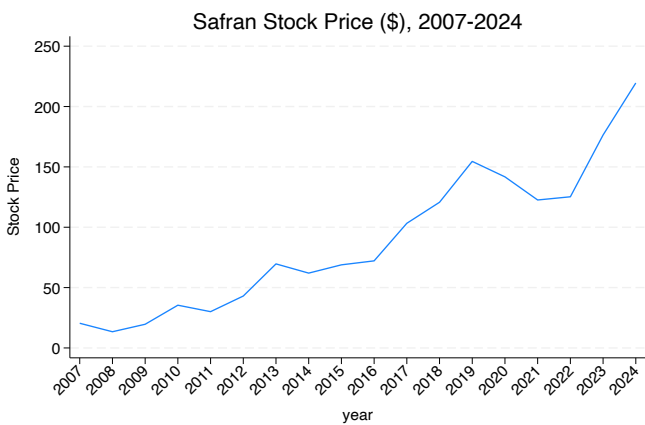
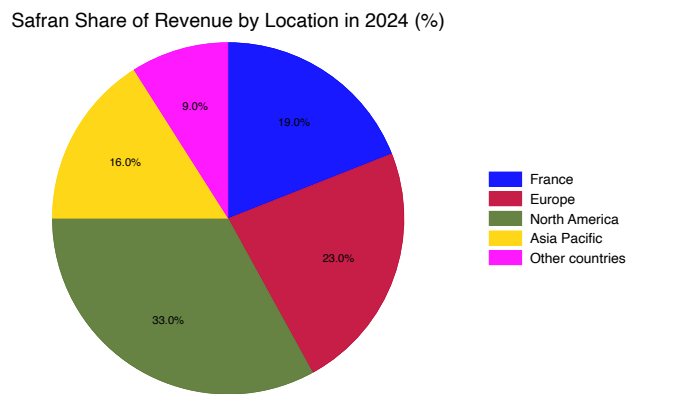
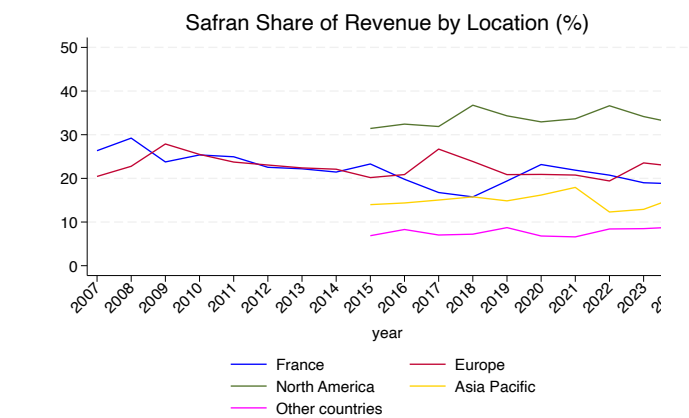
Schneider Electric



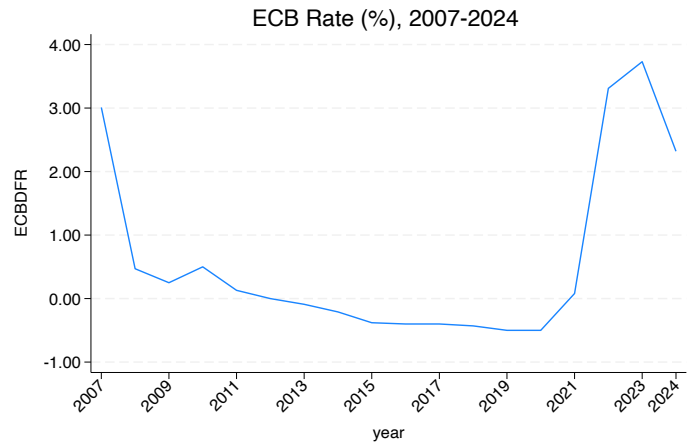
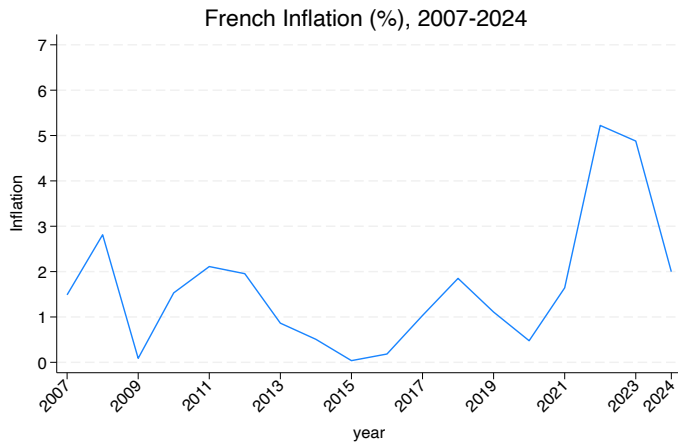
Schneider Electric Share of Revenue by Location in 2024 (%)



Safran



Appendix 2: French Inflation and European Central Bank rates



Appendix 3: PRINCIPAL COMPONENT ANALYSIS – Asian Markets

Principal components/correlation	Number of obs	=	378
	Number of comp.	=	4
	Trace	=	4
Rotation: (unrotated = principal)	Rho	=	1.0000

Component	Eigenvalue	Difference	Proportion	Cumulative
Comp1	2.919	2.118	0.730	0.730
Comp2	0.801	0.591	0.200	0.930
Comp3	0.211	0.142	0.053	0.983
Comp4	0.069	.	0.017	1.000

Principal components (eigenvectors)

Variable	Comp1	Comp2	Comp3	Comp4	Unexplained
NIFTY	0.533	-0.255	0.724	-0.356	0
ShanghaiCo~e	0.338	0.906	0.165	0.192	0
MSCIAsiaex~n	0.553	0.023	-0.653	-0.518	0
NIKKEI	0.545	-0.336	-0.148	0.754	0

Appendix 4: Correlations between Markets

Correlations between International market Indices

Variables	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
(1) CAC40Index	1.000							
(2) SP500	0.749	1.000						
(3) NIKKEI	0.738	0.932	1.000					
(4) ShanghaiCompos~e	0.572	0.209	0.298	1.000				
(5) NIFTY	0.850	0.944	0.875	0.361	1.000			
(6) MSCIEmergingMa~s	0.624	0.384	0.527	0.594	0.531	1.000		
(7) MSCIASiaexclJa~n	0.723	0.727	0.866	0.532	0.768	0.863	1.000	
(8) VIX	-0.358	-0.375	-0.427	-0.294	-0.408	-0.511	-0.535	1.000

CAC 40 Correlation with European Indicators

Variables	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
(1) CAC40Index	1.000							
(2) DAXIndex	0.815	1.000						
(3) UKIndex	0.584	0.406	1.000					
(4) EuropeanVIX	-0.607	-0.753	-0.608	1.000				
(5) MSCIEurope	0.949	0.849	0.753	-0.725	1.000			
(6) ECBDFR	0.615	0.344	0.295	-0.185	0.470	1.000		
(7) EURUSD	-0.270	-0.657	0.083	0.610	-0.297	-0.080	1.000	
(8) EUconsumerSent~t	0.142	0.252	0.290	-0.511	0.270	-0.452	-0.292	1.000

CAC 40 Correlation with European Indicators

Variables	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
(1) CAC40Index	1.000							
(2) smallCAC	0.718	1.000						
(3) MidandSmallCAC	0.699	0.979	1.000					
(4) french10yyield	0.032	-0.562	-0.604	1.000				
(5) FranceCDS	-0.595	-0.598	-0.603	0.292	1.000			
(6) GDPgrowth	0.170	0.142	0.103	0.133	0.125	1.000		
(7) Unemployment	-0.644	-0.225	-0.228	-0.325	0.244	-0.041	1.000	
(8) France_consume~t	0.315	0.569	0.447	-0.315	-0.397	0.274	0.084	1.000

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